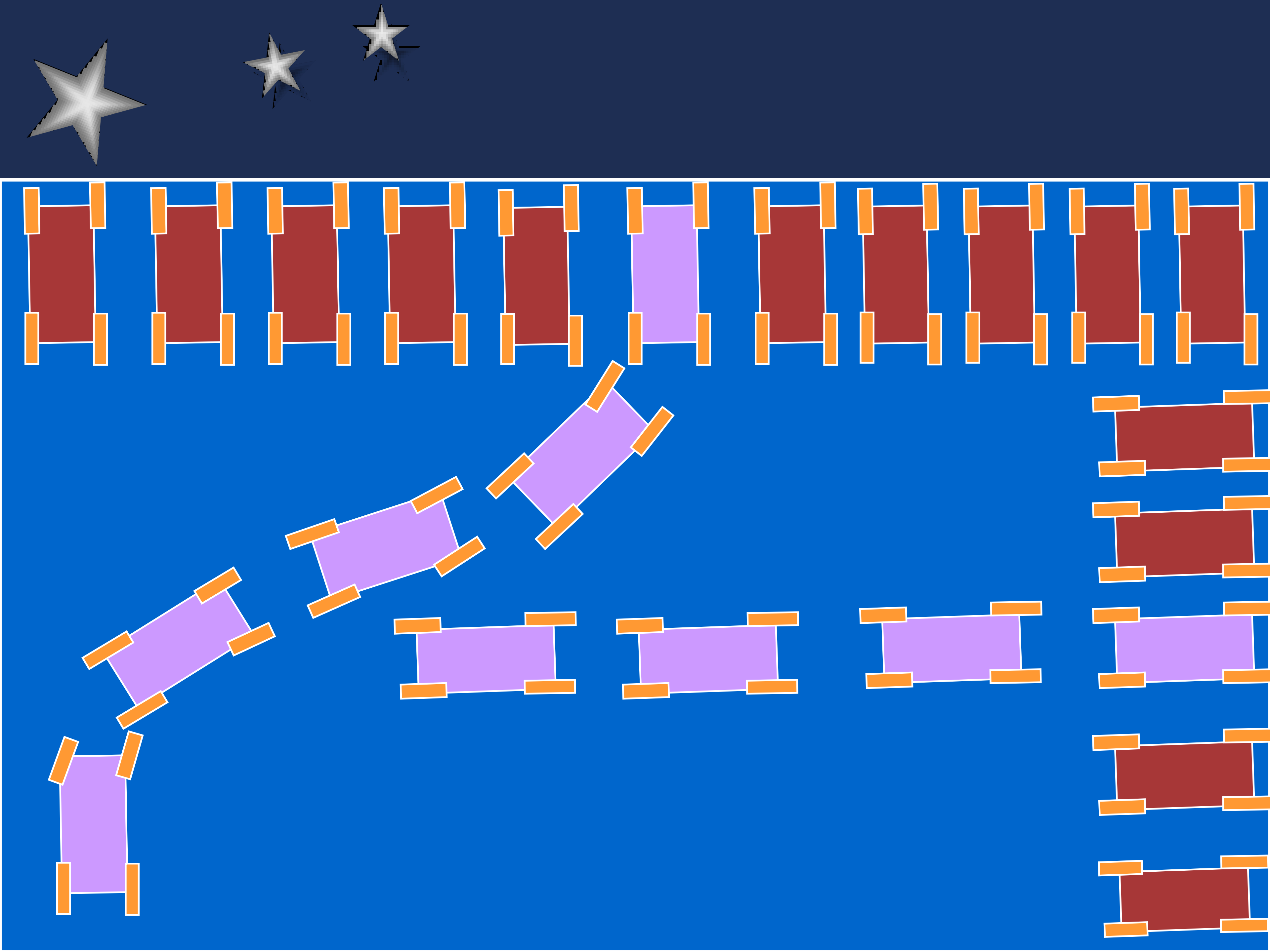
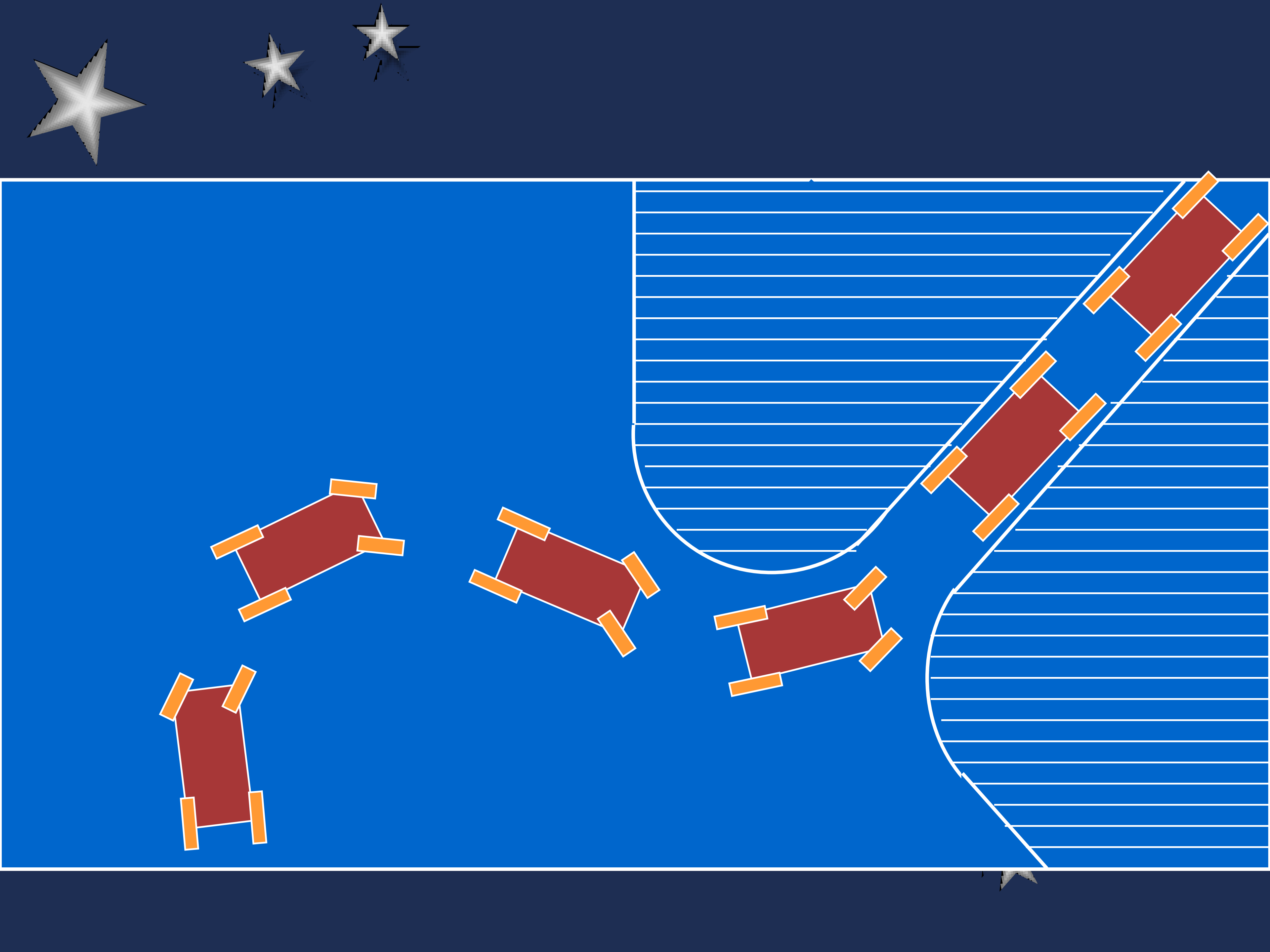
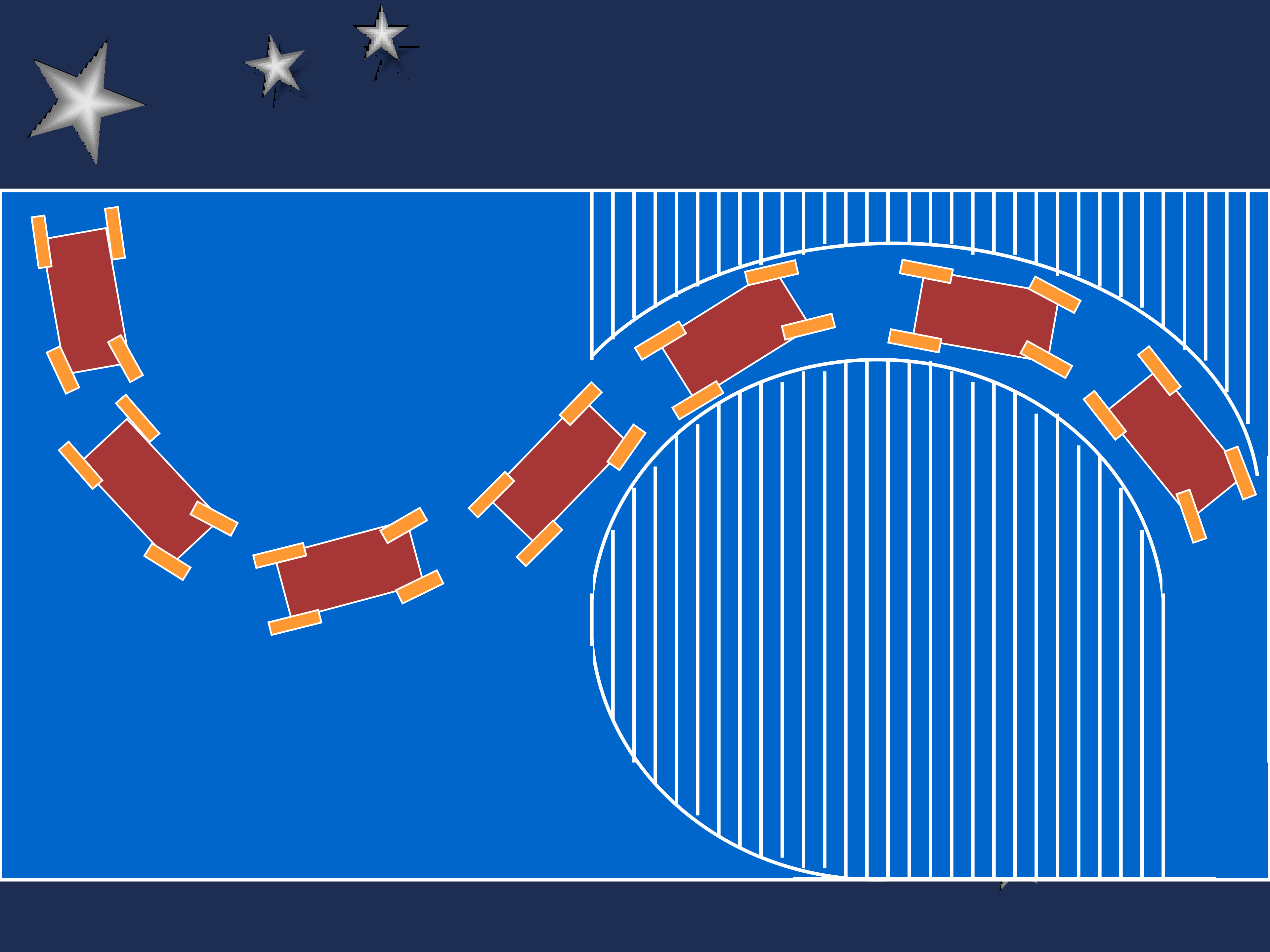
Control de Robots Móviles

Antonio Morán, Ph.D.

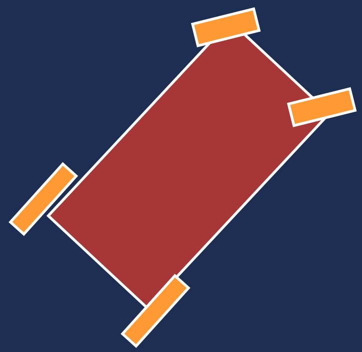






Tipos de Robots Móviles

Tipo Carro

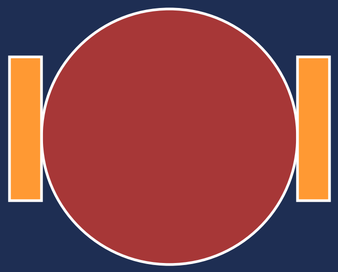


Cuatro ruedas

Ruedas delanteras: Dirección

Ruedas traseras: Tracción

Diferencial



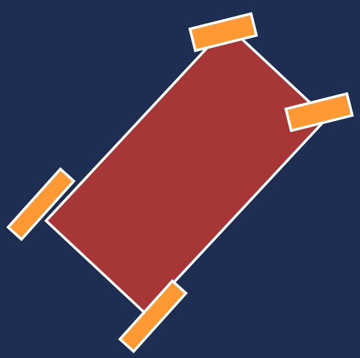
Dos ruedas

Las dos ruedas son para tracción y dirección

Tipos de Robots Móviles

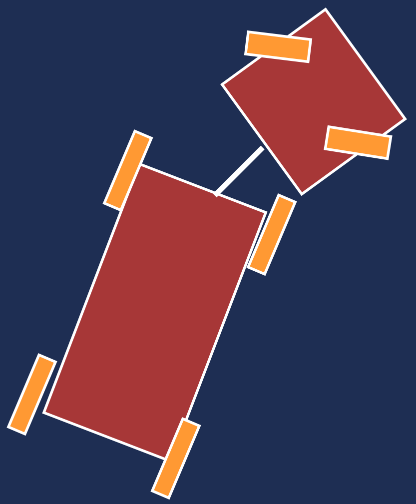
Tipo Carro

Tipo Carro



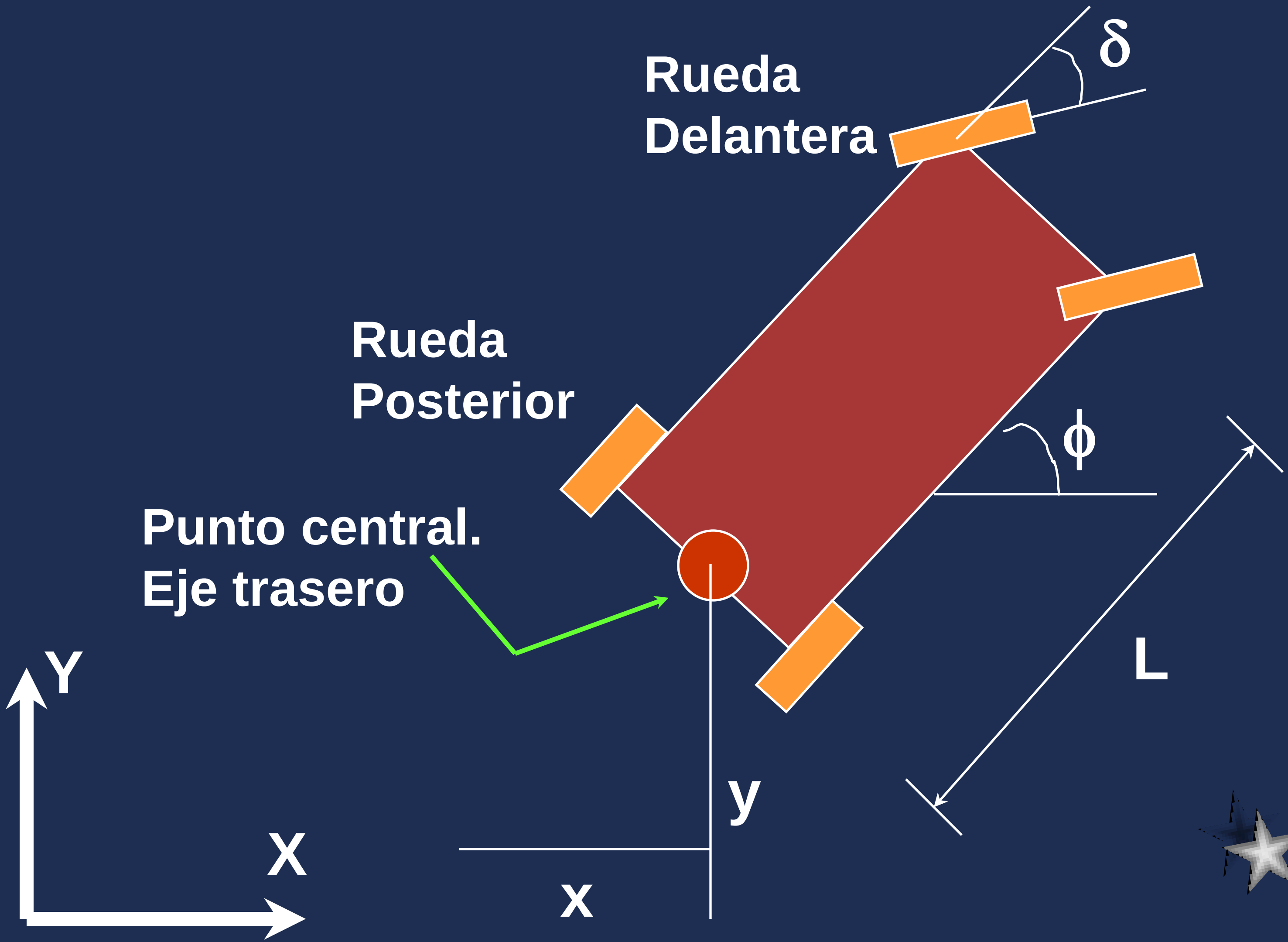
Un solo cuerpo

Tipo Trailer

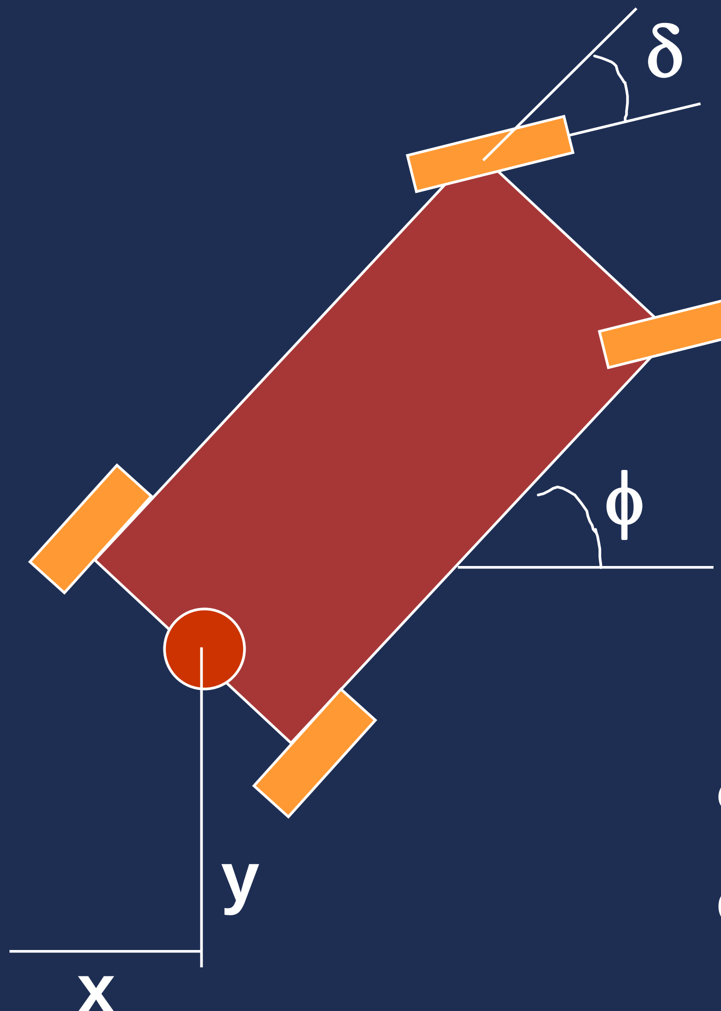


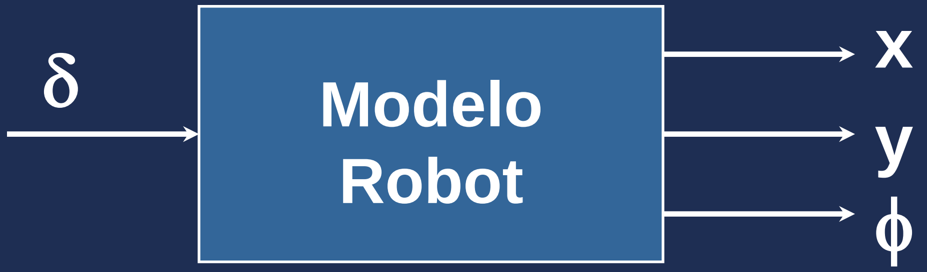
Dos cuerpos articulados

Modelamiento de Robots Móviles Tipo Carro



Modelamiento de Robots Móviles Tipo Carro





δ horario: Positivo

ϕ antihorario: Positivo

Modelamiento de Robot Móvil

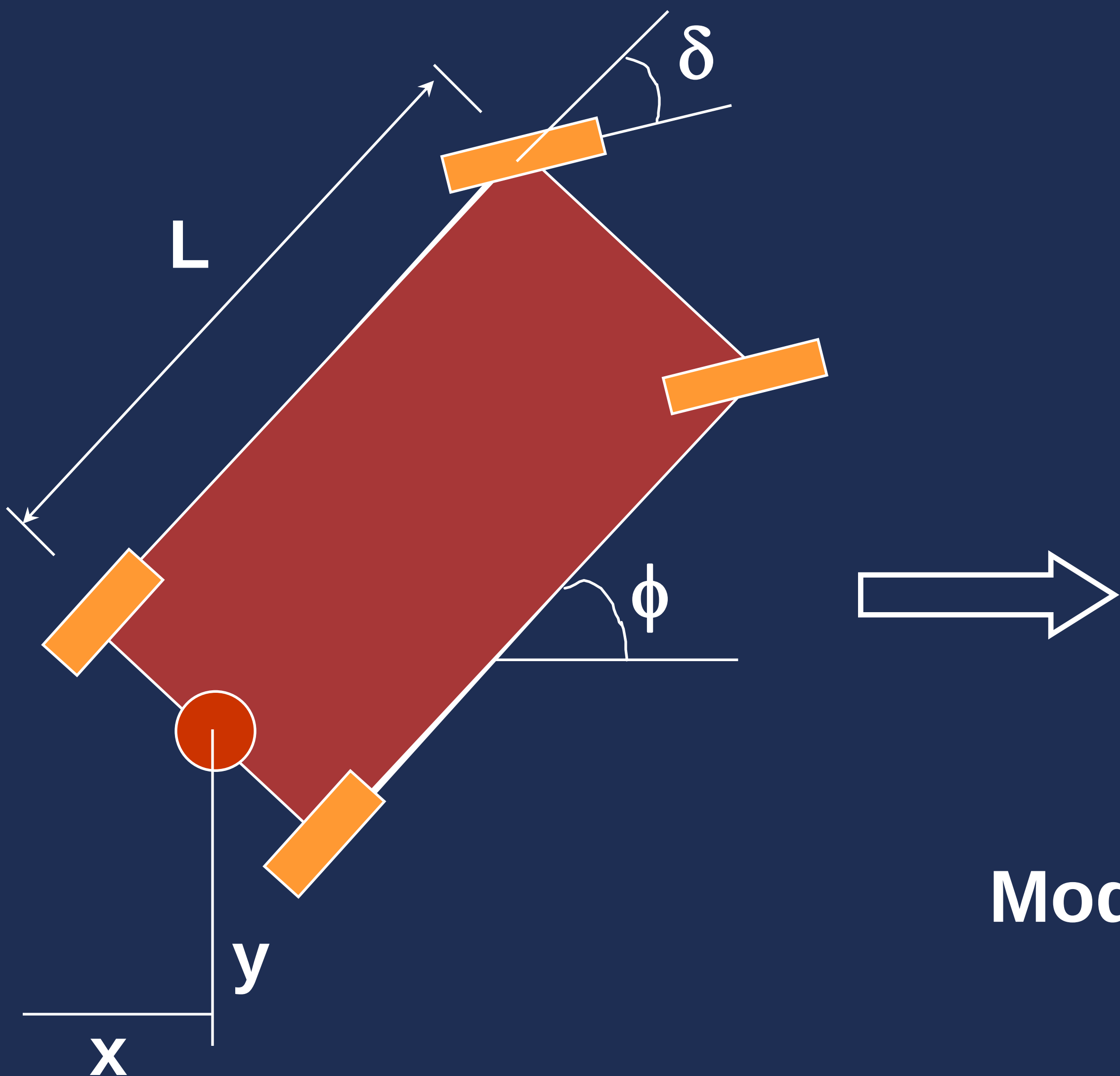
Dada la posición inicial x_o y_o ϕ_o

Dado el ángulo del timón δ

⇓

Se puede determinar la trayectoria del robot

Modelo de Dos Ruedas



Modelo Bicicleta

Modelamiento de Robots Móviles Tipo Carro

Asumpsiones:

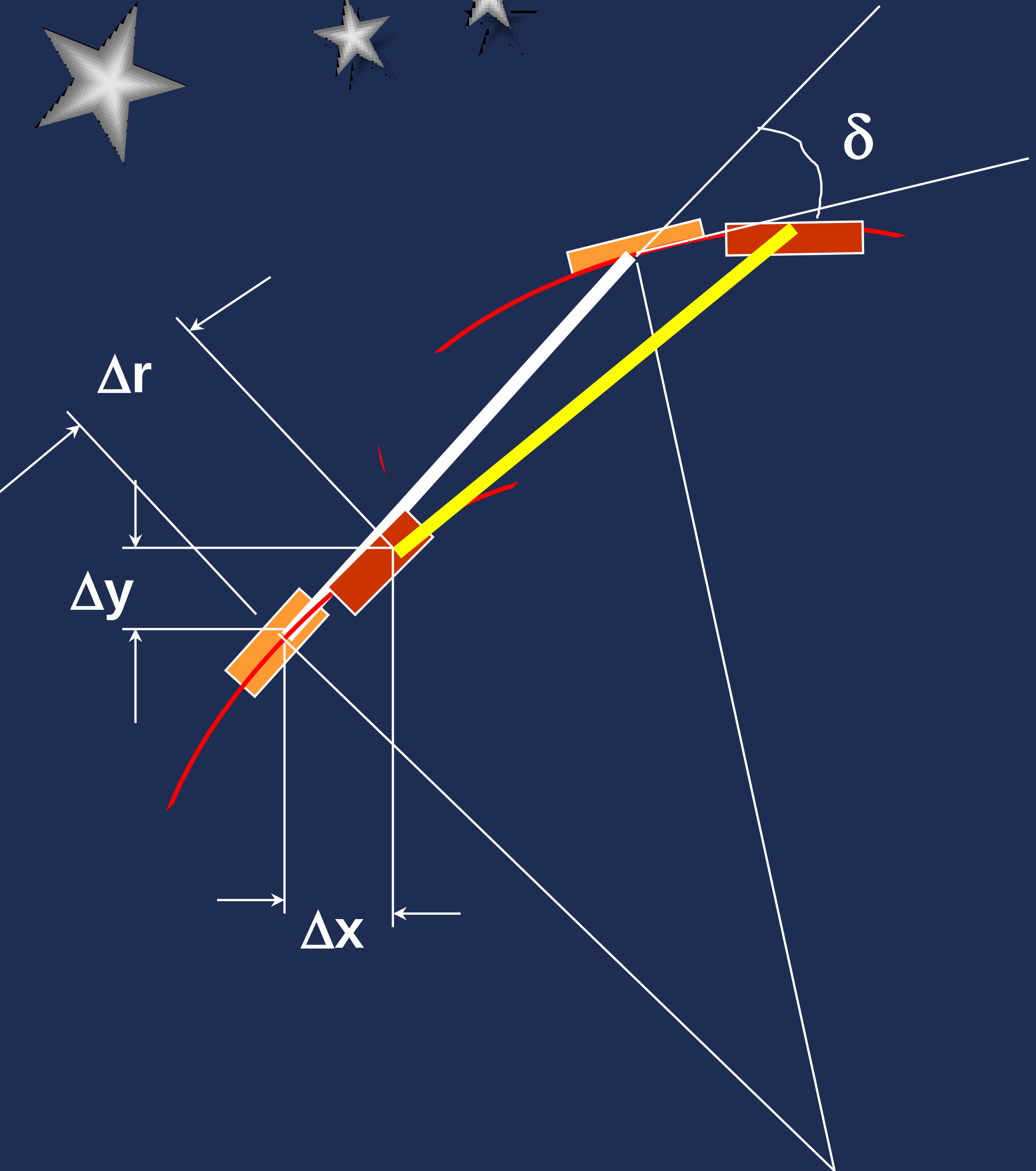
Velocidad baja << 3m/seg

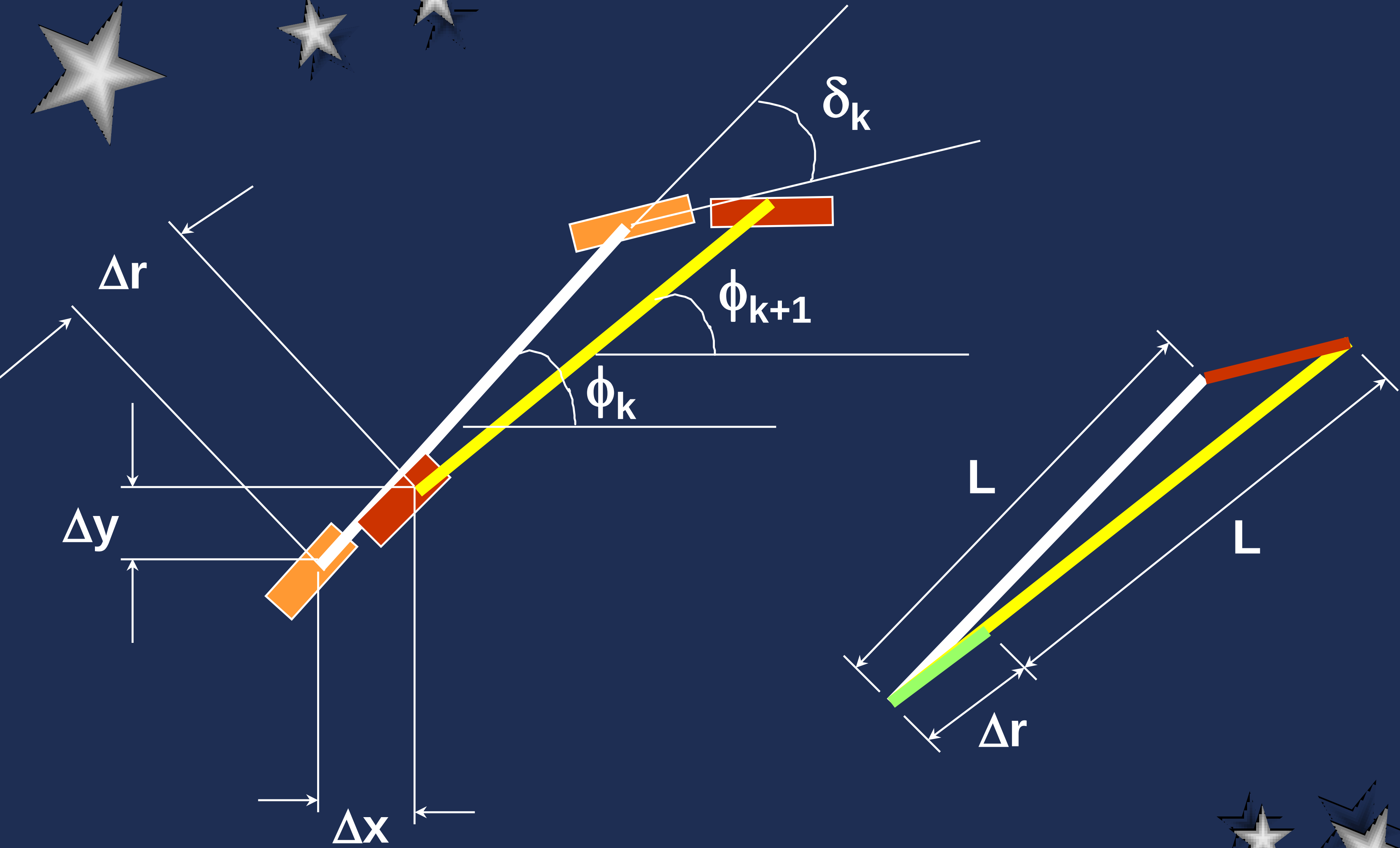
La ruedas no deslizan

Modelo cinemático

Velocidad ruedas traseras (tracción):

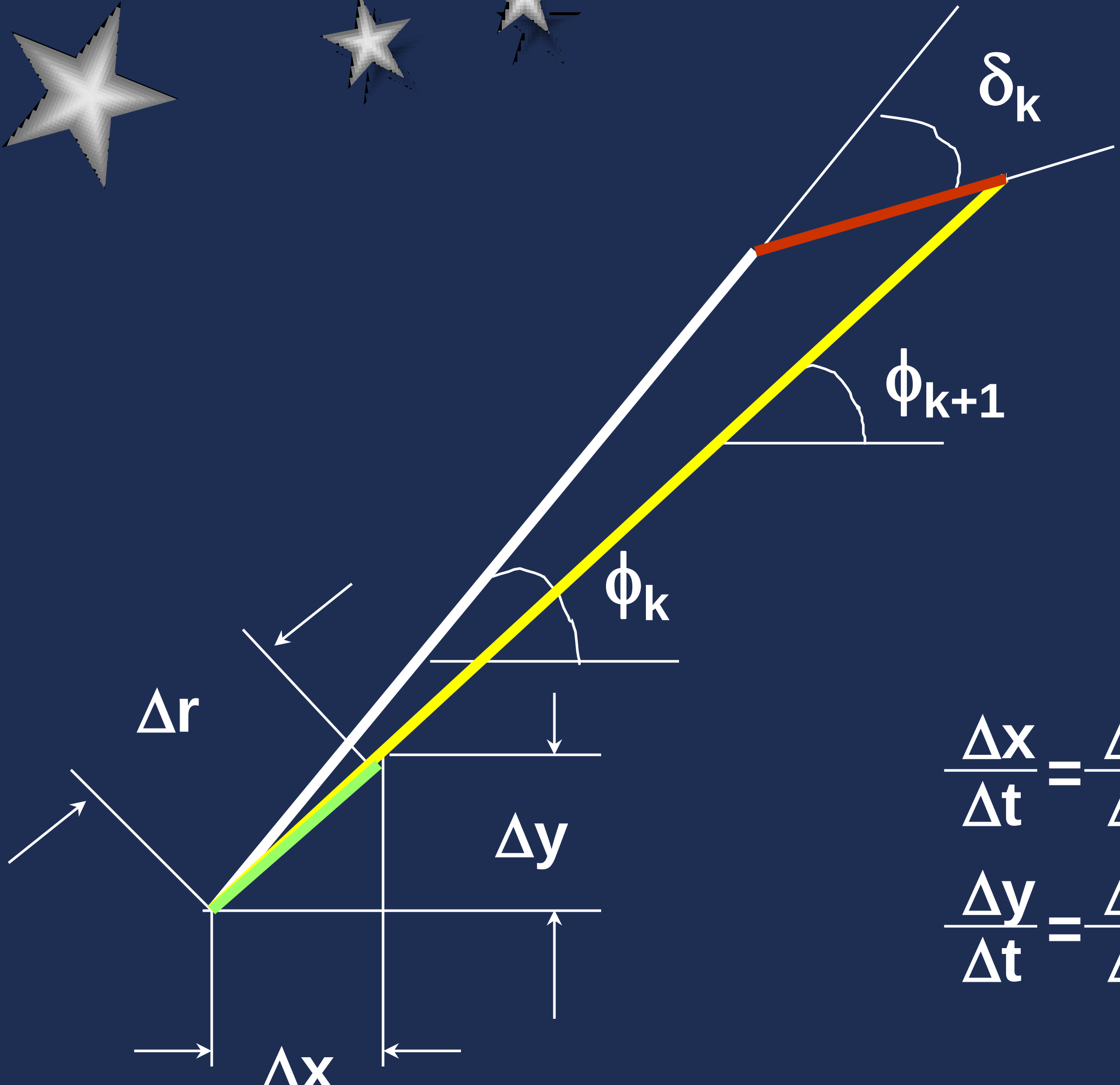
$$v = \frac{\Delta r}{\Delta t}$$

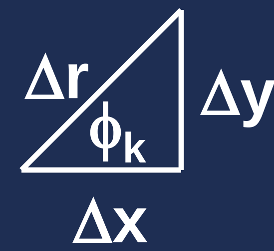

δ
Δr
Δy
Δx



$$x_{k+1} = x_k + \Delta x$$

$$y_{k+1} = y_k + \Delta y$$

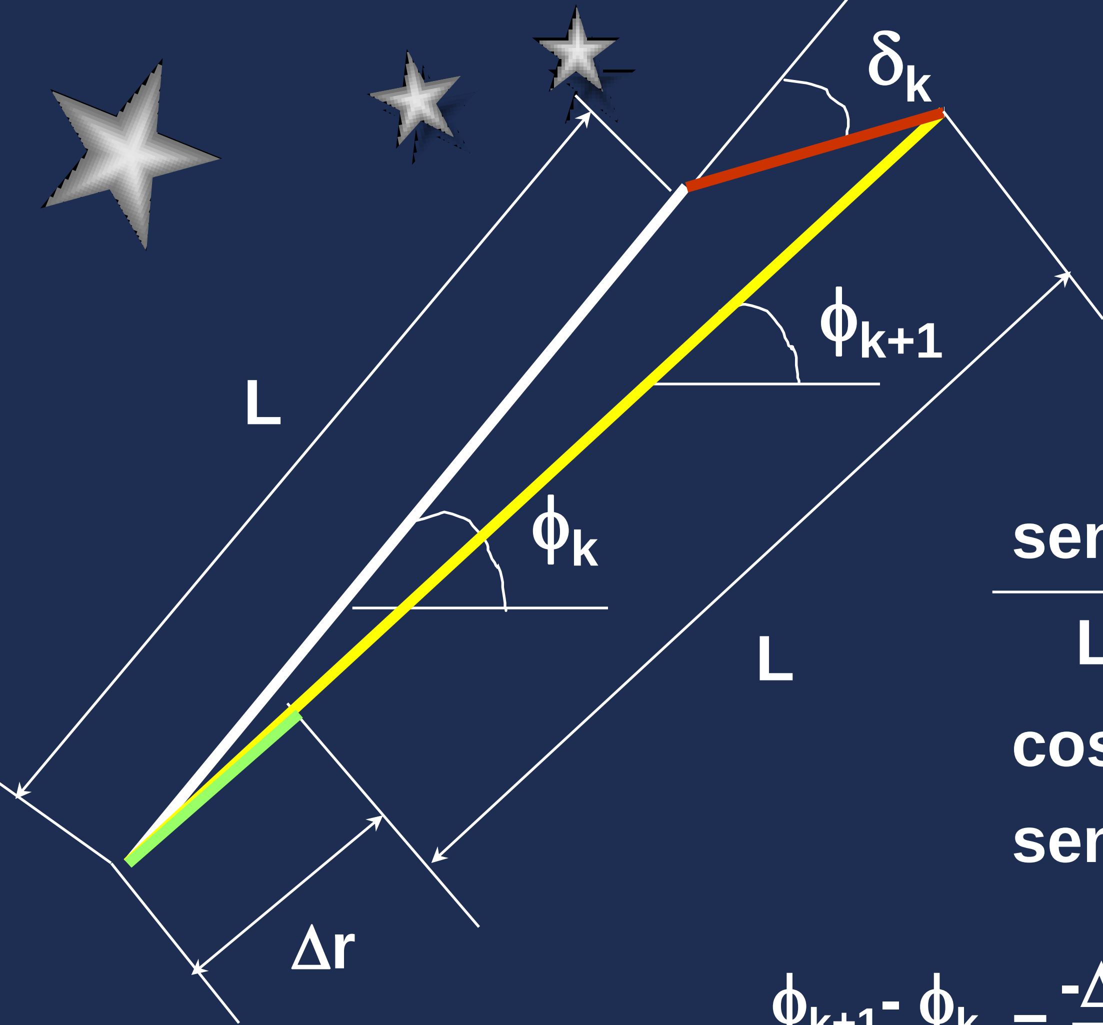




$$\Delta x = \Delta r \cos(\phi_k)$$
$$\Delta y = \Delta r \,\mathrm{sen}(\phi_k)$$

$$\frac{\Delta x}{\Delta t} = \frac{\Delta r}{\Delta t} \cos(\phi_k)$$
$$\frac{\Delta y}{\Delta t} = \frac{\Delta r}{\Delta t} \,\mathrm{sen}(\phi_k)$$

$$\dot{x} = v \cos(\phi)$$
$$\dot{y} = v \,\mathrm{sen}(\phi)$$



$$\frac{\text{sen}(\pi-\delta_k)}{L+\Delta r} = \frac{\text{sen}(\delta_k - \phi_{k+1} + \phi_k)}{L}$$

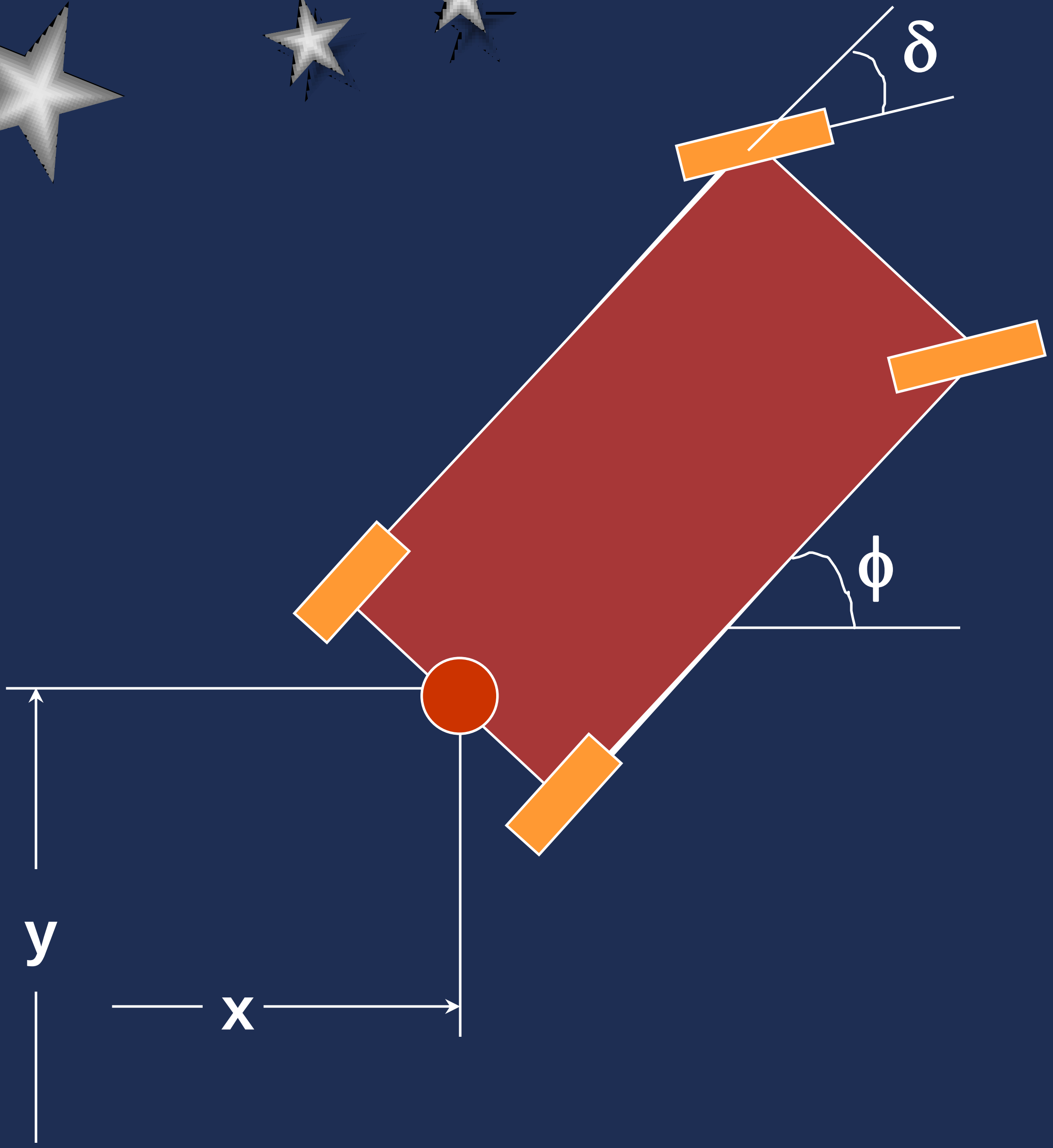
$$\cos(\phi_{k+1} - \phi_k) \approx 1$$

$$\text{sen}(\phi_{k+1} - \phi_k) \approx \phi_{k+1} - \phi_k$$

$$\phi_{k+1} - \phi_k = \frac{-\Delta r \tan(\delta_k)}{L+\Delta r}$$

$L >> \Delta r$

$$\dot{\phi} = \frac{-v}{L} \tan(\delta)$$



Modelo del Robot

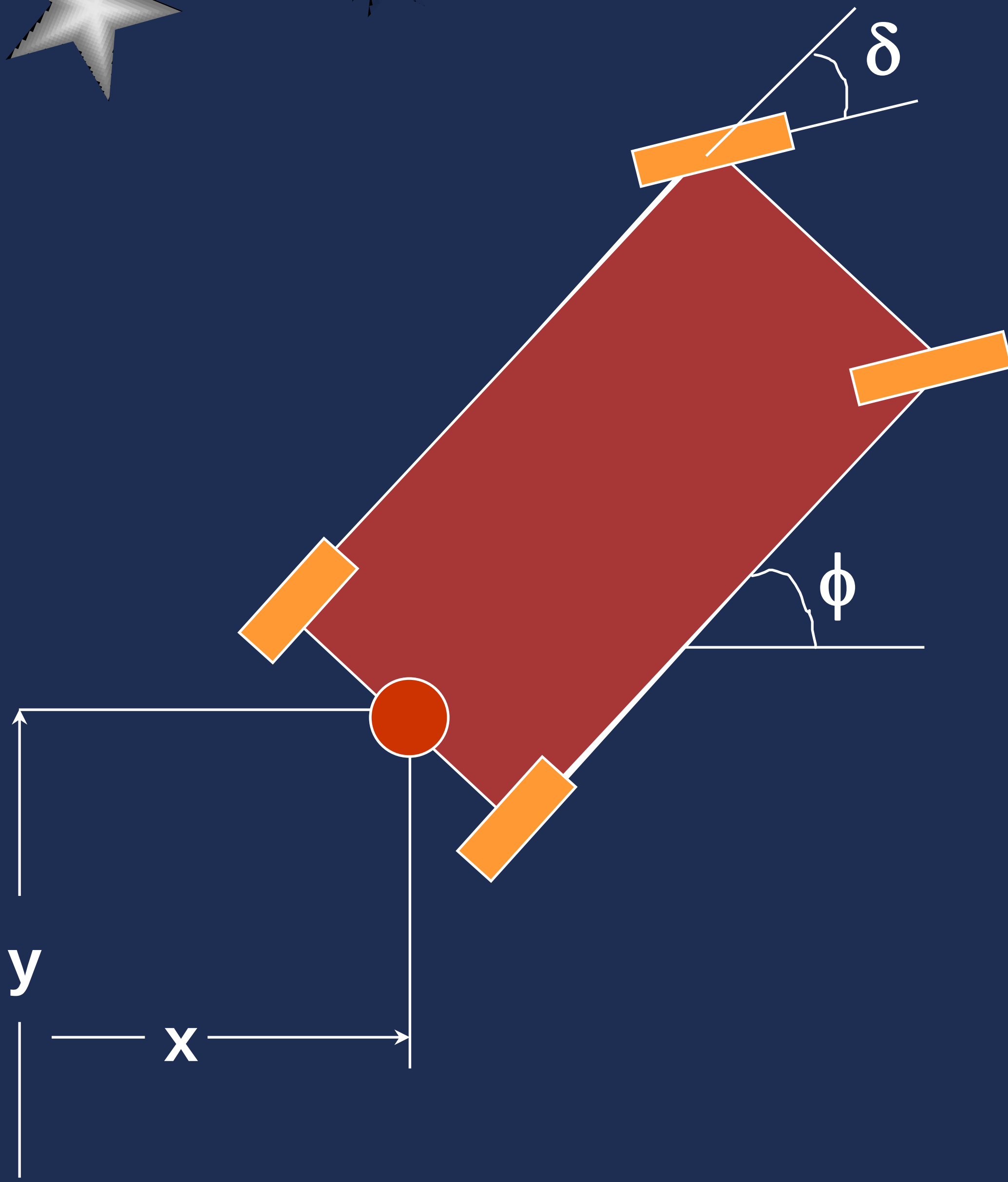
Ecuaciones de Movimiento

$$\dot{x} = v\cos(\phi)$$

$$\dot{y} = v\,\mathrm{sen}(\phi)$$

$$\dot{\phi} = \frac{-v}{L}\tan(\delta)$$

Ecuaciones de Movimiento



$$x_{k+1} = x_k + \Delta r \cos(\phi_k)$$

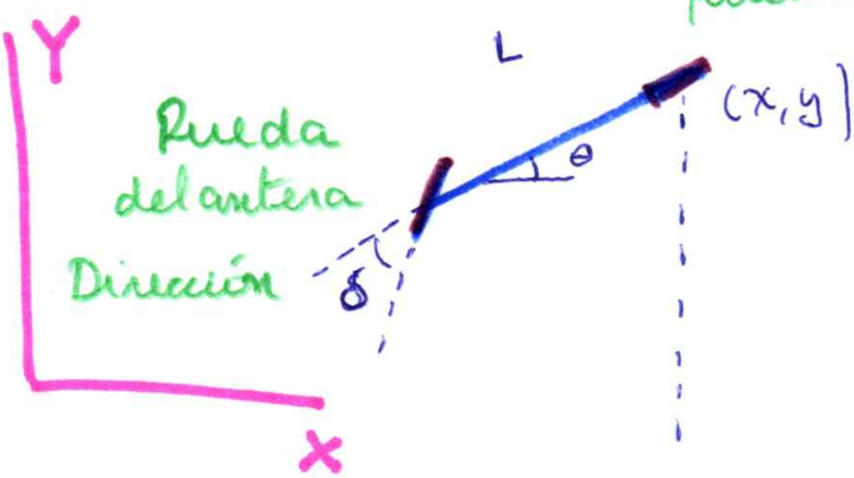
$$y_{k+1} = y_k + \Delta r \sin(\phi_k)$$

$$\phi_{k+1} = \phi_k - \frac{\Delta r}{L} \tan(\delta)$$

$$v = \frac{\Delta r}{\Delta t}$$

Modelamiento de Robot Móvil Tipo Carro

Rueda posterior de tracción



velocidad de la rueda de traccion = v

$$v = \frac{T}{\Delta t}$$

Asunciones:
- Baja velocidad $V < 3\ m/seg$ constant
- No resbala, no desliza
- Movimiento hacia atrás $(V > 0)$

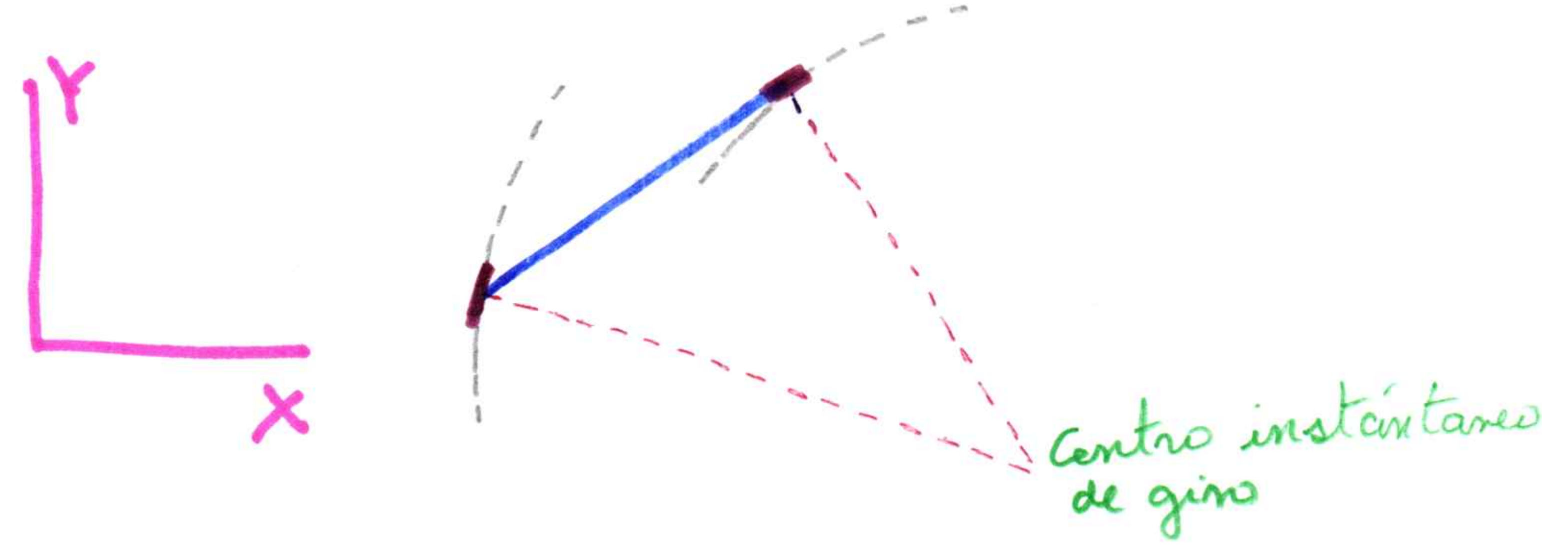
Modelamiento

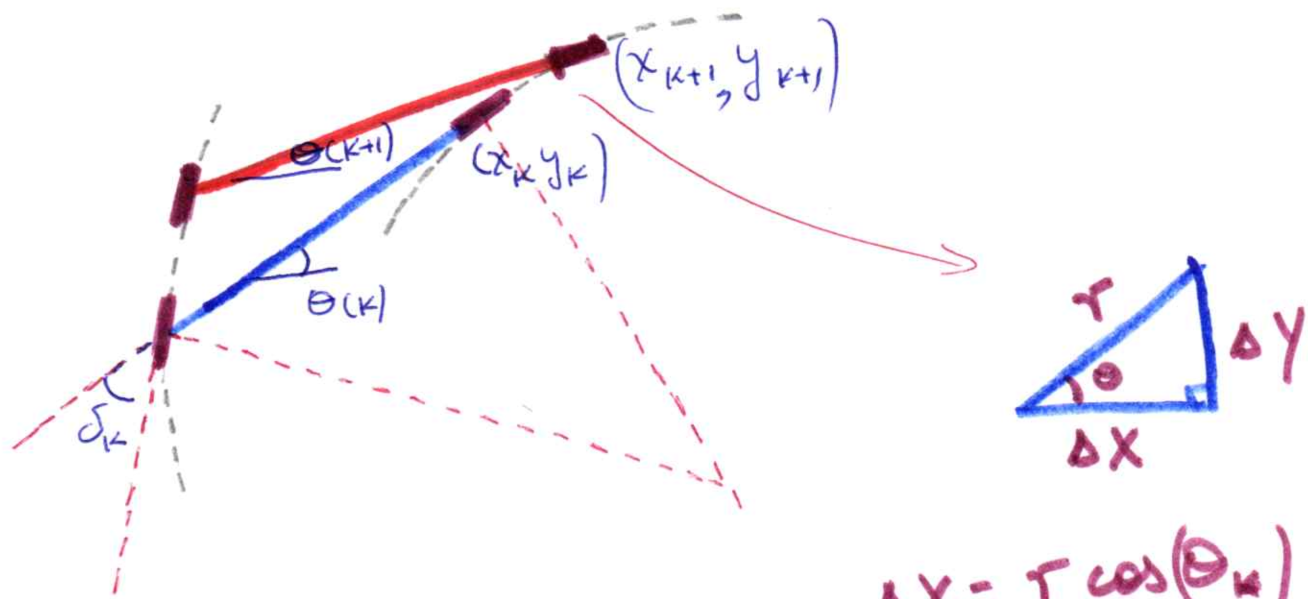
$$\dot{x} =$$

$$\dot{y} =$$

$$\dot{\theta} =$$

Analizando el movimiento en dos instantes consecutivos (K) y (K+1)





$$\Delta X = r\cos(\theta_K)$$

$$\Delta Y = r\,\text{sen}(\theta_K)$$

$$x_{K+1} - x_K = r\cos(\theta(K))$$

$$\frac{x_{K+1} - x_K}{\Delta t} = \frac{r}{\Delta t}\cos(\theta(K))$$

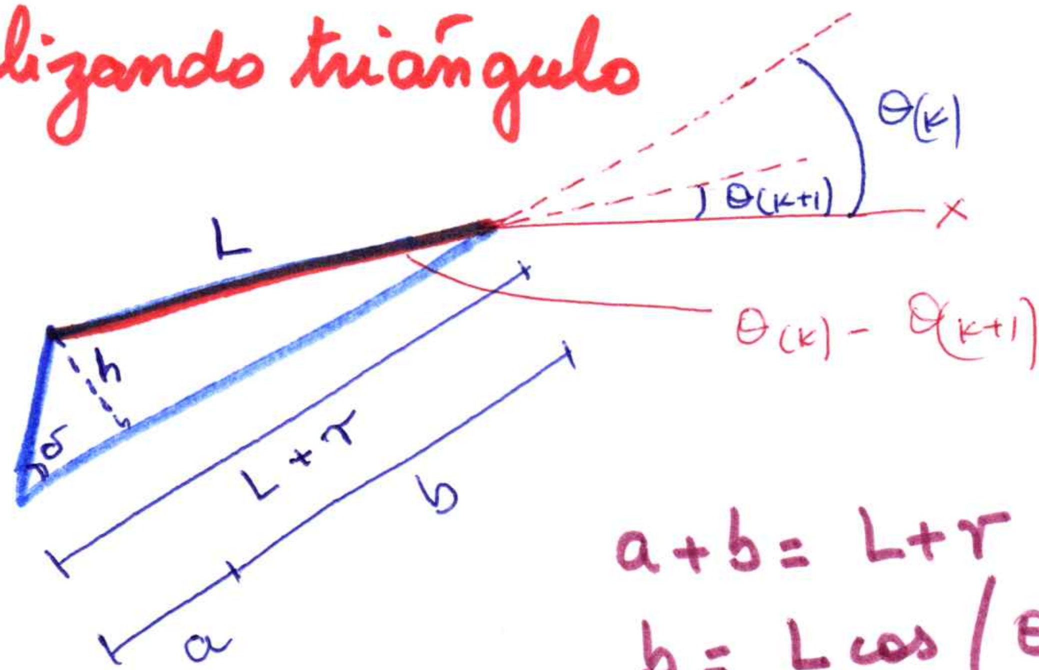
$$\boxed{\dot{x} = v\cos(\theta)}$$

$$y_{k+1} - y_k = r \, sen(\theta_{(k)})$$

$$\frac{y_{k+1} - y_k}{\Delta t} = \frac{r}{\Delta t} sen(\theta_{(k)})$$

$$\boxed{\dot{y} = v \, sen(\theta)}$$

Analizando triángulo



$$a + b = L + r$$

$$b = L \cos\left(\theta_{(k)} - \theta_{(k+1)}\right)$$

$$h = L \, sen\left(\theta_{(k)} - \theta_{(k+1)}\right)$$

$$\tan(\delta_{(k)}) = \frac{h}{a}$$

$$a = \frac{h}{\tan(\delta_k)} = \frac{L \, sen(\theta_{(k)} - \theta_{(k+1)})}{\tan(\delta_k)}$$

Δt es pequeño $\Rightarrow$ $\theta_{(k)} - \theta_{(k+1)} \approx 0$

$$a+b = L+r$$

$$\frac{L\,\underbrace{\operatorname{sen}(\theta_{(k)}-\theta_{(k+1)})}_{\theta_{(k)}-\theta_{(k+1)}}}{\tan(\delta_k)} + L\underbrace{\cos(\theta_{(k)}-\theta_{(k+1)})}_{1} = L+r$$

$$\frac{L\left(\theta_{(k)}-\theta_{(k+1)}\right)}{\tan(\delta_k)} + L = L+r$$

$$\frac{\theta_{(k+1)}-\theta_{(k)}}{\Delta t} = \frac{-r}{\Delta t\, L}\tan(\delta_k)$$

$$\dot{\theta} = -\frac{v}{L}\tan(\delta)$$

Modelo del robot móvil tipo carro:

$$\dot{x} = v\cos(\phi)$$

$$\dot{y} = v\operatorname{sen}(\phi)$$

$$\dot{\phi} = -\frac{v}{L}\tan(\delta)$$

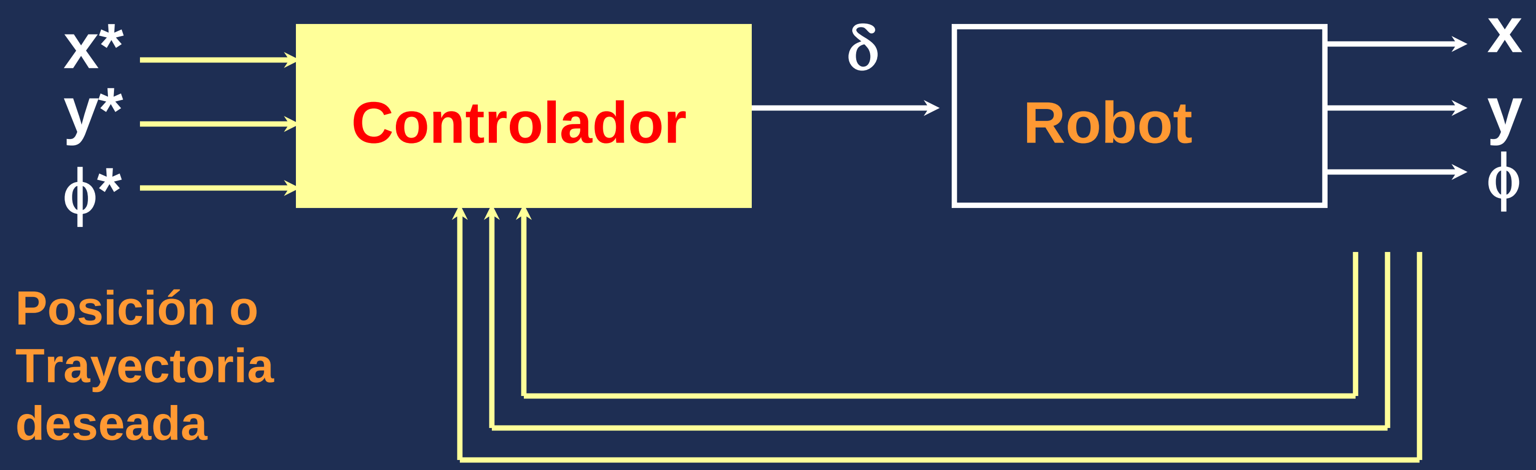
Control de Robots Móviles

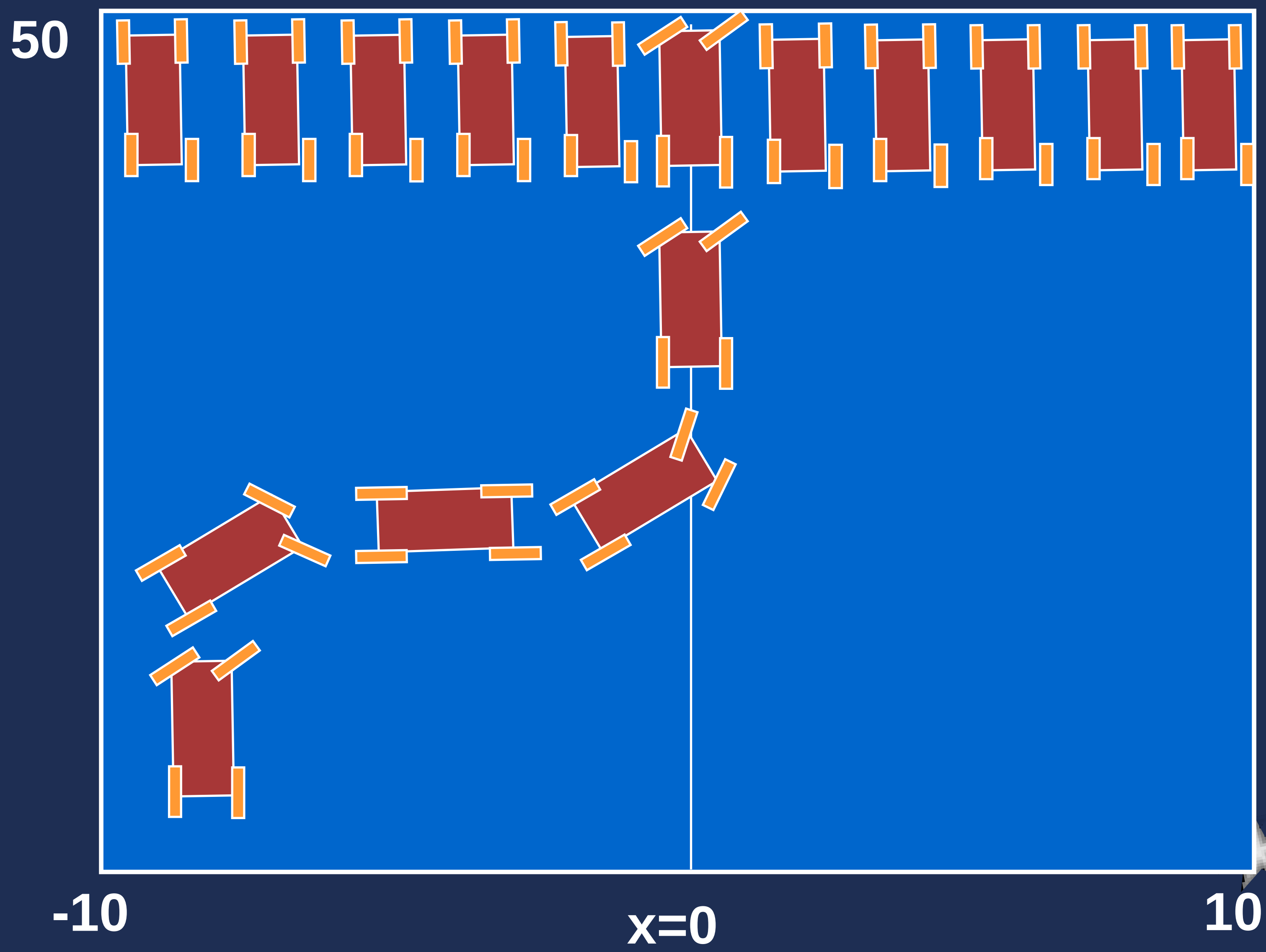
Como determinar el ángulo del timón δ de manera que el robot:

Alcance una posición determinada.

Siga una trayectoria deseada.

Control de Robots Móviles

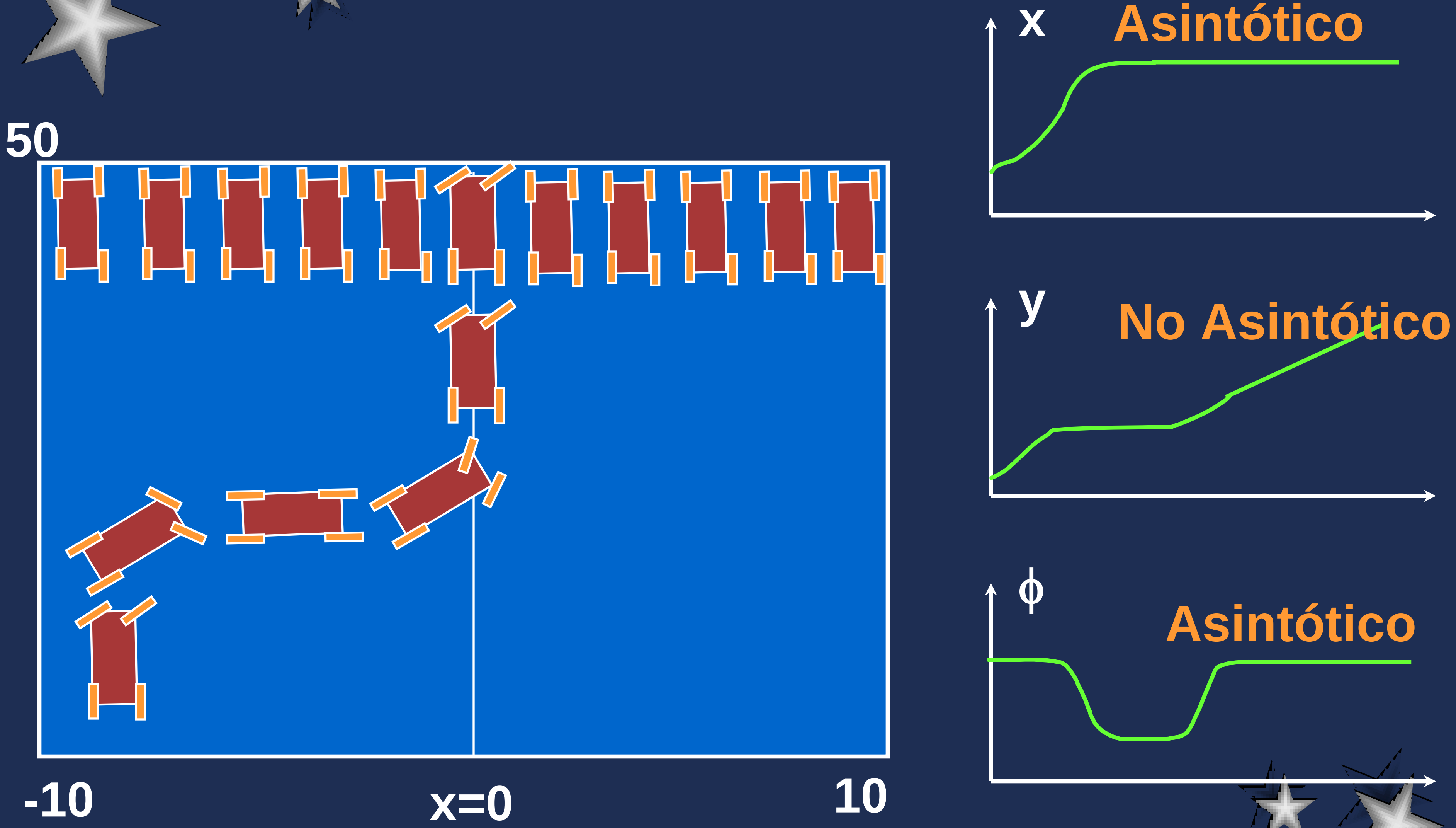




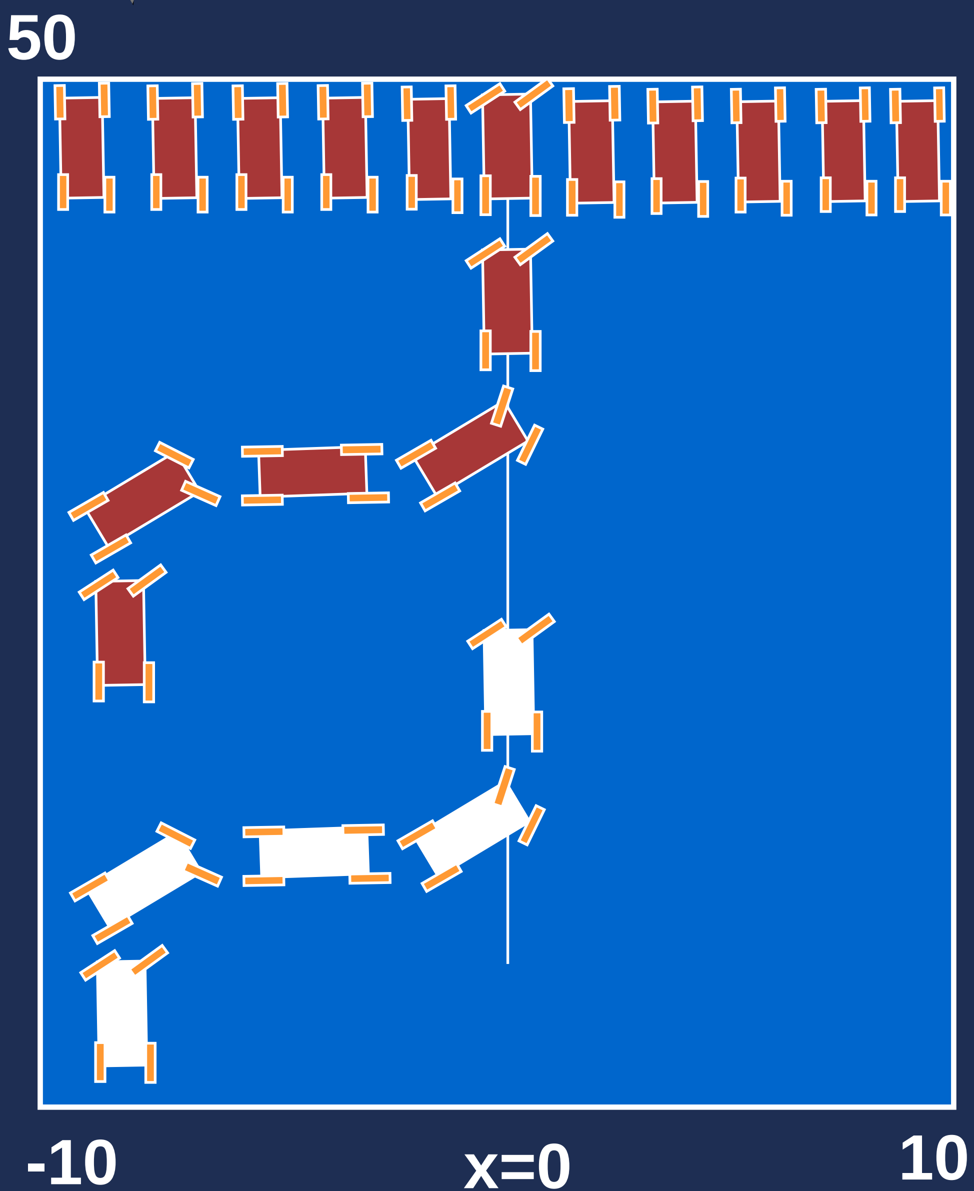
x* = 0

y* = 50

ϕ* = π/2


50
-10
x=0
10
x
Asintótico
y
No Asintótico
ϕ
Asintótico

Control Strategy

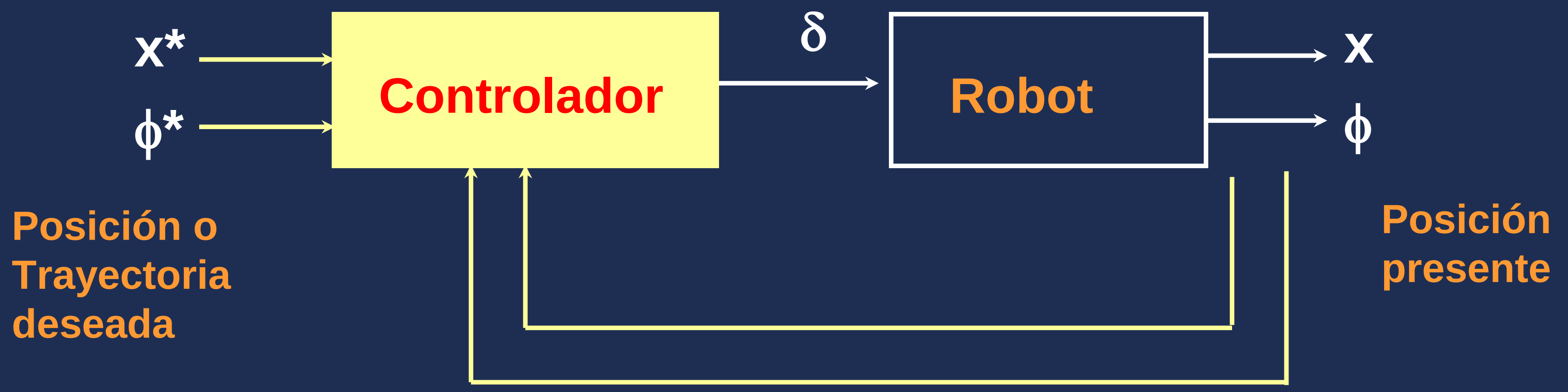


Para evitar la colisión con los vehículos estacionados, el carro primero llegará a la recta x = 0 y después continuará defrente a la posición deseada

Dos posiciones inciales distintas: igual x_o y ϕ_o peero diferente y_o

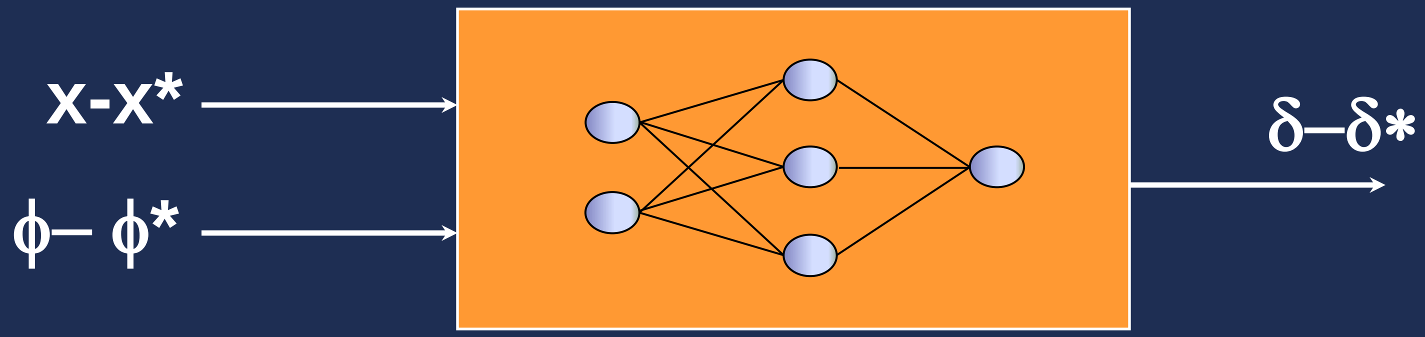
La señal de control (ángulo del timón δ) no depende de y

Control de Robots Móviles



La señal de control (ángulo del timón δ) no depende de y

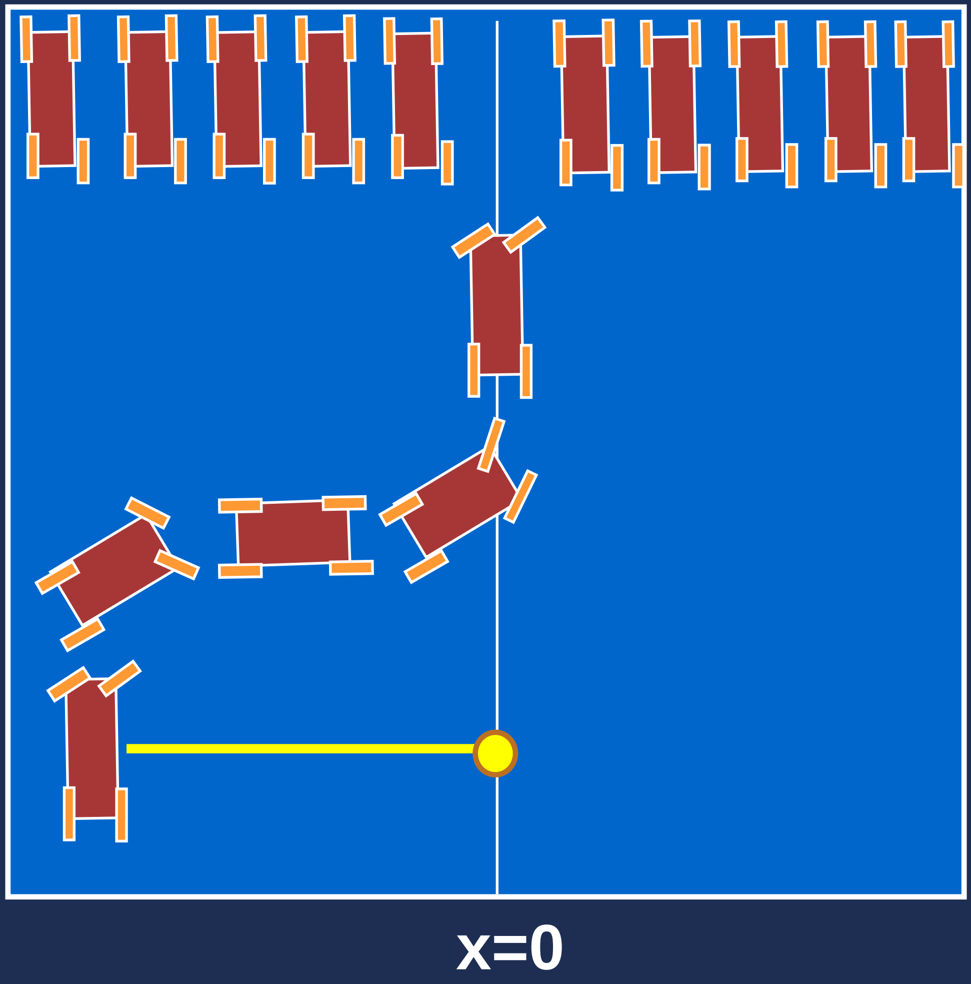
Neuro-Controlador



$x^* = 0$

$\phi^* = \pi/2$

$\delta^* = 0$



Objetivo

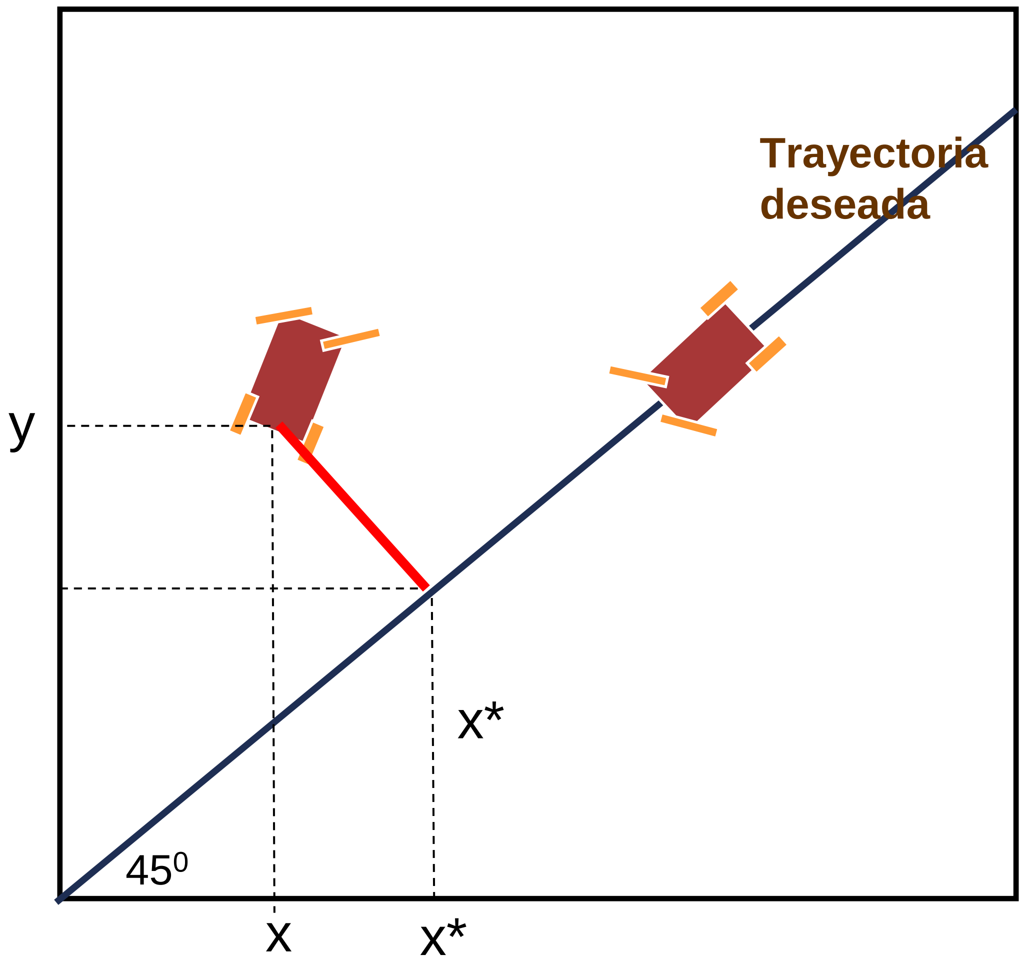
$x^* = 0$

$\phi^* = \pi/2$

Cómo se define la posición deseada en cada instante:

Trazar una perpendicular a la recta deseada, el punto de intersección corresponde a la coordenada x deseada y la pendiente al ϕ deseado

Seguimiento de Trayectoria Recta 45^0



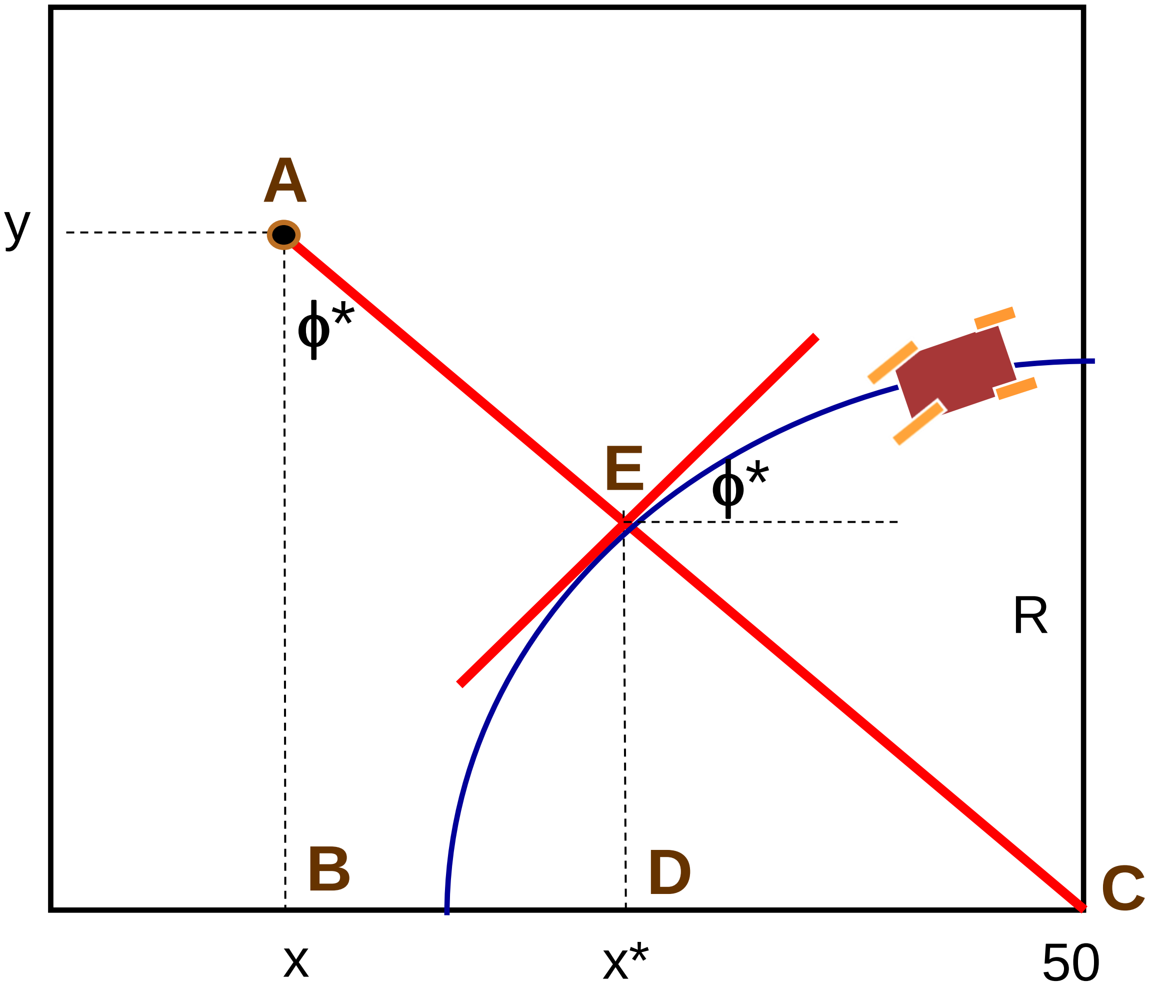
$x^*+(x^*-x) = y$

$x^* = (x+y)/2$

$\phi^* = \pi/4$

$\delta^* = 0$

Seguimiento de Trayectoria Circular R



Triángulos ABC y EDC semejantes

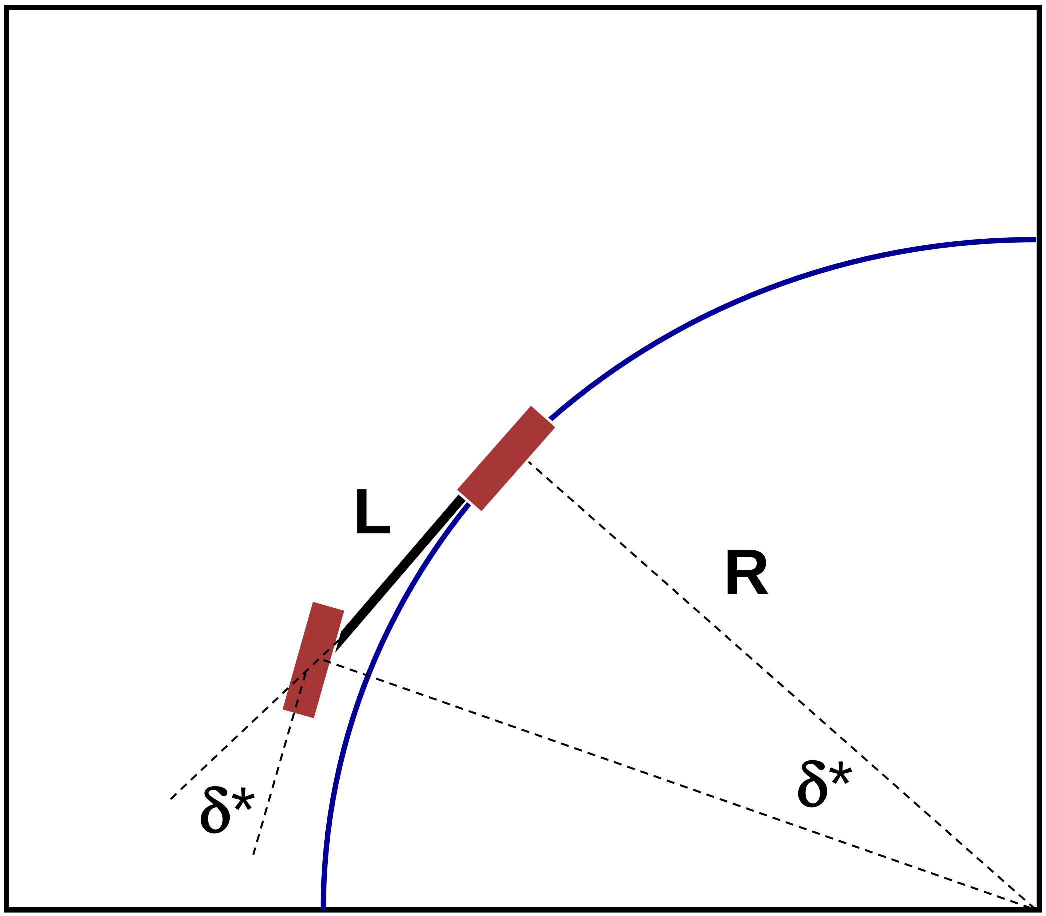
$$\frac{AC}{EC} = \frac{BC}{DC}$$

x* =

ϕ* = atan((50-x)/y)

δ* =

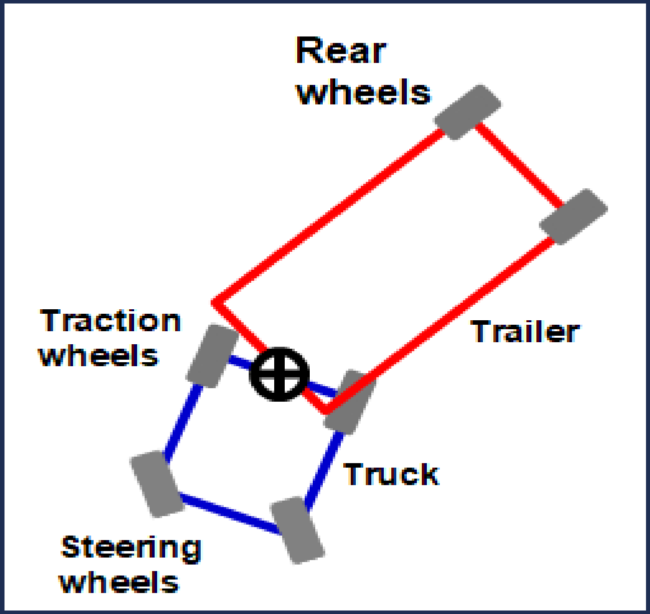
Seguimiento de Trayectoria Circular R



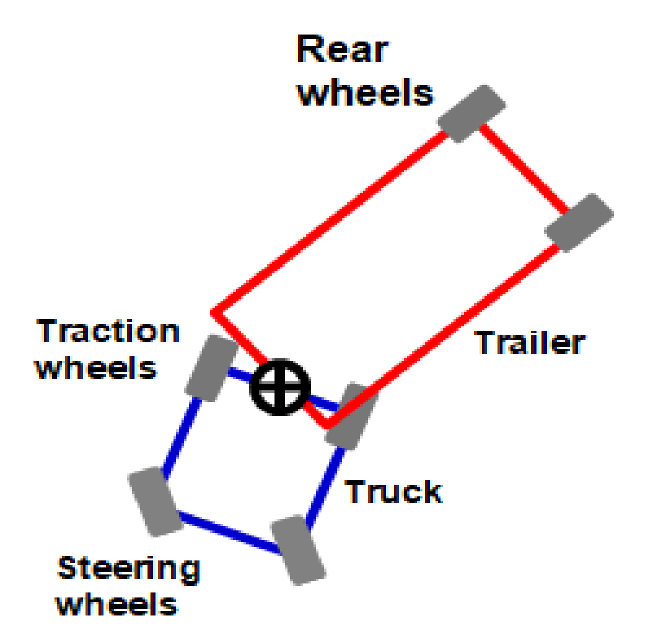
$$\delta^* = \text{atan}(L/R)$$

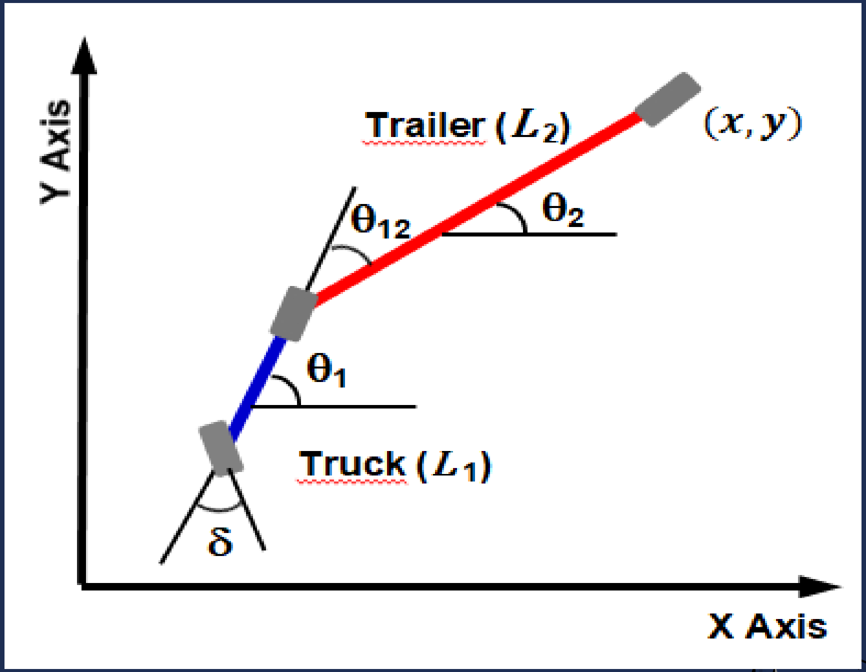
Modelamiento de Robot Móvil Tipo Trailer

Modelamiento de Robot Móvil Tipo Trailer

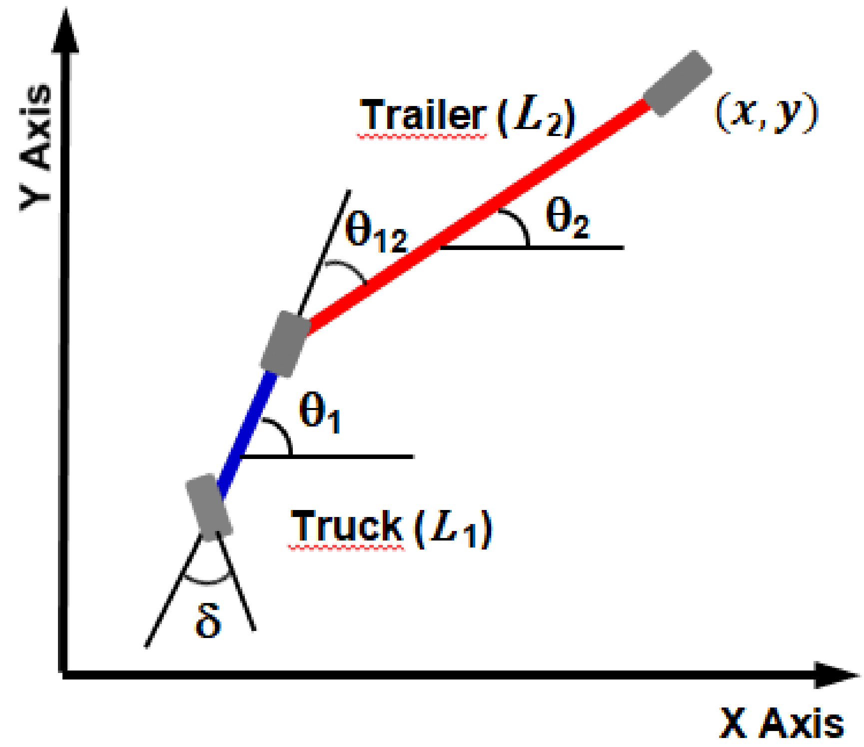


Modelamiento de Robot Móvil Tipo Trailer





Modelamiento de Robot Móvil Tipo Trailer



$$\dot{x} = v \cos\theta_{12} \cos\theta_2$$

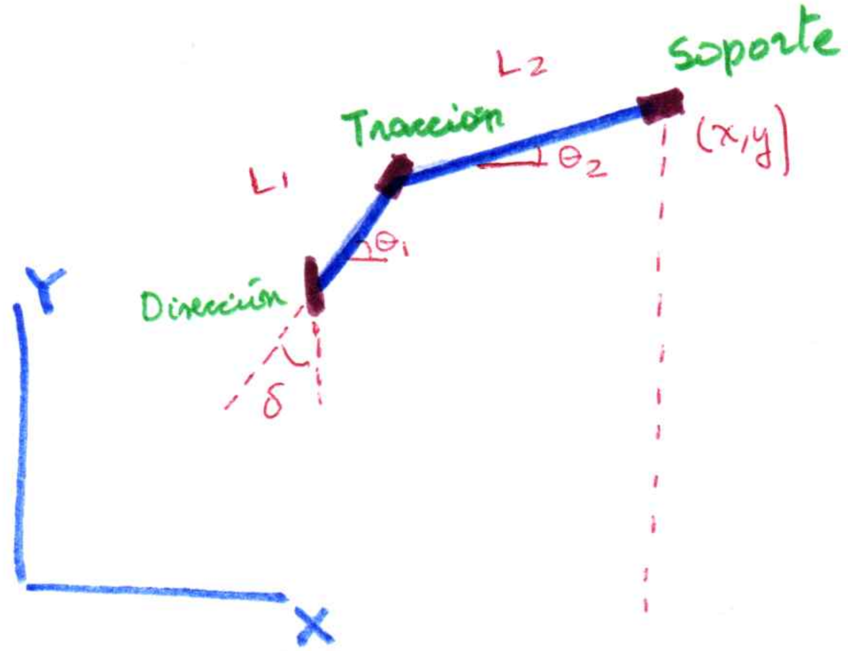
$$\dot{y} = v \cos\theta_{12} \sin\theta_2$$

$$\dot{\theta}_1 = -\frac{v}{L_1} \tan\delta$$

$$\dot{\theta}_2 = -\frac{v}{L_2} \sin\theta_{12}$$

$$\dot{\theta}_{12} = \frac{v}{L_2} \sin\theta_{12} - \frac{v}{L_1} \tan\delta$$

Modelamiento de Truck-Trailer

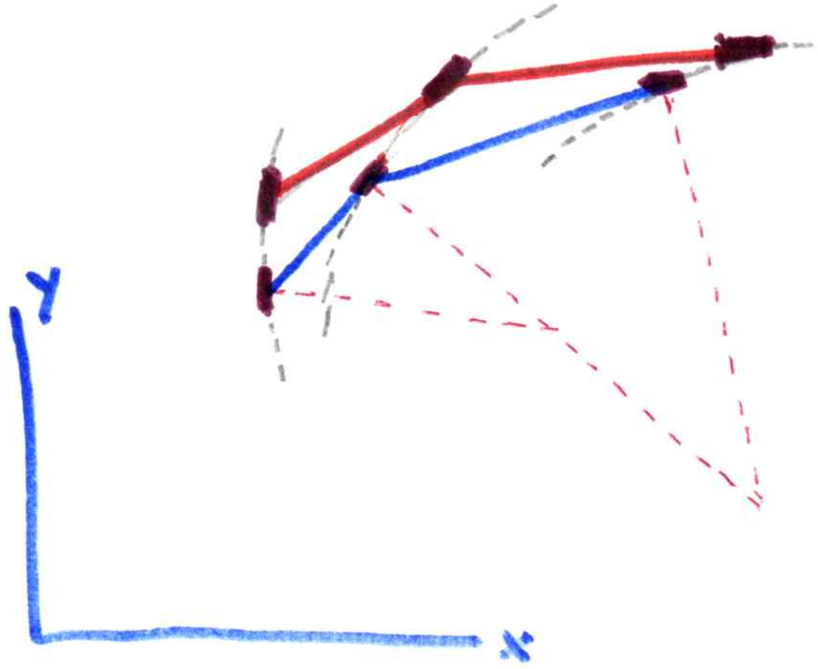


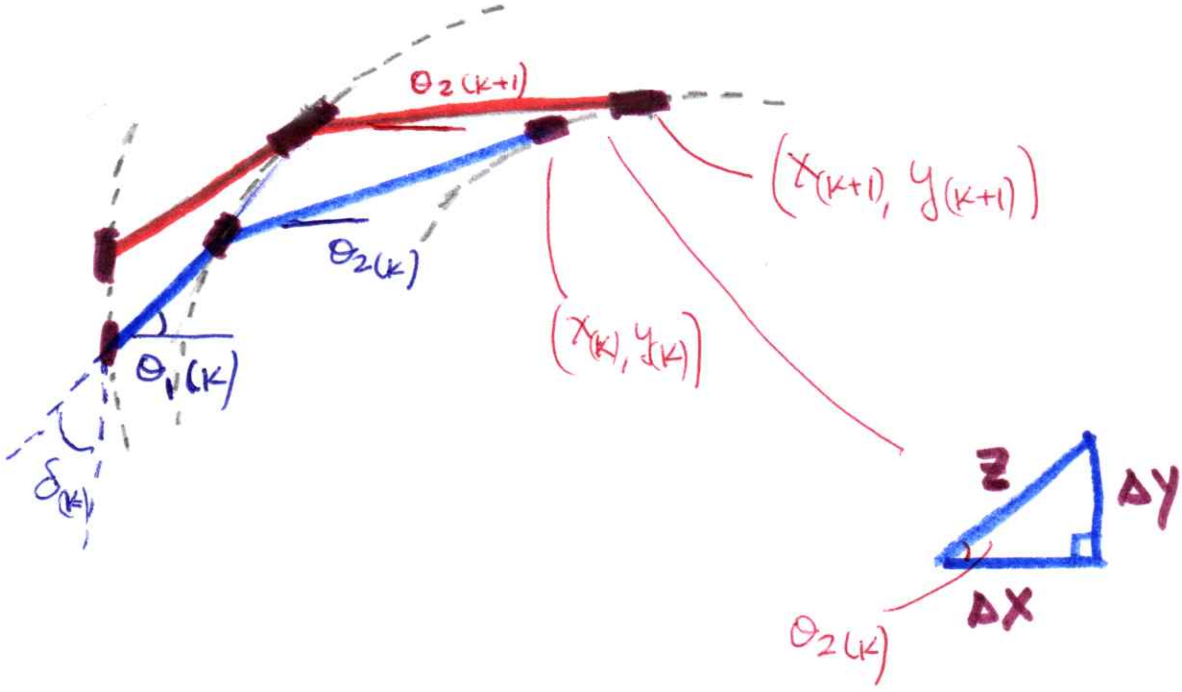
Las ruedas de la tracción se desplazan con una velocidad v

$$v = \frac{r}{\Delta t}$$

Analizando la posición del robot móvil para dos instantes de tiempo (k) y $(k+1)$

↳ avance de la rueda de la tracción $= r$





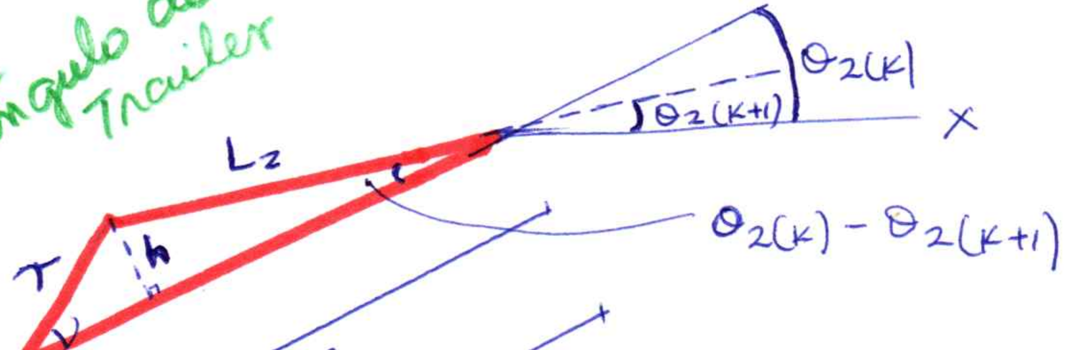
$$\Delta X = Z\cos(\theta_2(k))$$

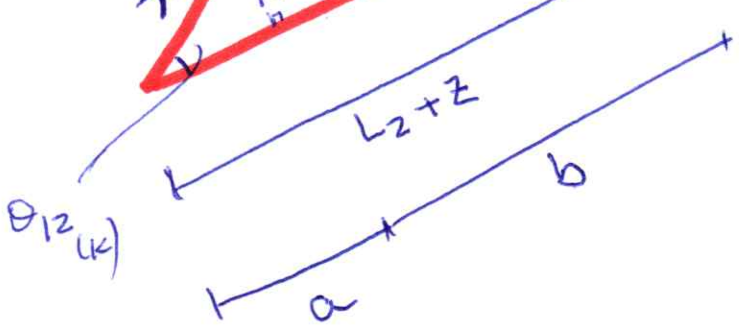
$$\Delta Y = Z\,\mathrm{sen}(\theta_2(k))$$

$$X_{(k+1)} - X_{(k)} = Z\cos(\theta_2(k))$$

$$y(k+1) - y(k) = Z\,\mathrm{sen}(\theta_2(k))$$

Triángulo del Trailer





$$a + b = L_2 + Z$$

$$b = L_2\cos(\theta_2(k) - \theta_2(k+1))$$

$$h = L_2\,\mathrm{sen}(\theta_2(k) - \theta_2(k+1))$$

$$\theta_{12} = \theta_1 - \theta_2$$

$$a = T\cos\left(\theta_{12}(k)\right)$$

$$h = T\,\mathrm{sen}\left(\theta_{12}(k)\right)$$

$$L_2\,\mathrm{sen}\left(\underbrace{\theta_{2}(k) - \theta_{2}(k+1)}_{\theta_2(k) - \theta_2(k+1)\ \rightarrow\ \Delta\theta_2(k) \simeq 0}\right) = T\,\mathrm{sen}\left(\theta_{12}(k)\right)$$

$$L_2\left(\theta_2(k) - \theta_2(k+1)\right) = T\,\mathrm{sen}\left(\theta_{12}(k)\right)$$

$$\theta_2(k+1) - \theta_2(k) = -\frac{T}{L_2}\,\mathrm{sen}\left(\theta_{12}(k)\right)$$

$$\boxed{\dot{\theta}_2 = -\frac{v}{L_2}\,\mathrm{sen}(\theta_{12})}$$

$$a + b = L_2 + z$$

$$T\cos\left(\theta_{12}(k)\right) + L_2\underbrace{\cos\left(\theta_2(k) - \theta_2(k+1)\right)}_{1} = L_2 + z$$

$$T\cos\left(\theta_{12}(k)\right) + L_2 = L_2 + z$$

$$\boxed{z = T\cos(\theta_{12})}$$

Reemplazamos en las ecuaciones de $X(k)$, $Y(k)$

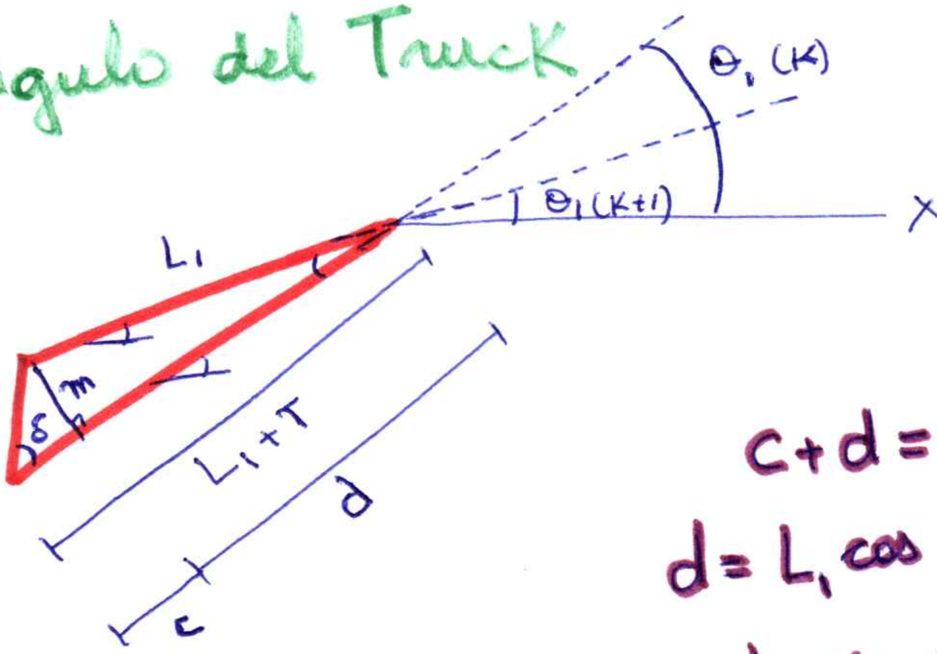
$$x_{(k+1)} - x_{(k)} = T \cos(\theta_{12(k)}) \cos(\theta_{2(k)})$$

$$\boxed{\dot{x} = v \cos(\theta_{12}) \cos(\theta_2)}$$

$$y_{(k+1)} - y_{(k)} = T \cos(\theta_{12(k)}) \,\mathrm{sen}(\theta_{2(k)})$$

$$\boxed{\dot{y} = v \cos(\theta_{12}) \,\mathrm{sen}(\theta_2)}$$

Triángulo del Truck



$$c + d = L_1 + T$$

$$d = L_1 \cos(\theta_{1(k)} - \theta_{1(k+1)})$$

$$m = L_1 \,\mathrm{sen}(\theta_{1(k)} - \theta_{1(k+1)})$$

$$\tan(\delta_{(k)}) = \frac{m}{c}$$

$$c = \frac{m}{\tan(\delta_{(k)})}$$

$$c = \frac{L_1 \,\mathrm{sen}(\theta_{1(k)} - \theta_{1(k+1)})}{\tan(\delta_k)}$$

$$c + d = L_1 + r$$

$$\frac{L_1 \operatorname{sen}(\theta_{1(k)} - \theta_{1(k+1)})}{\tan(\delta_k)} + L_1 \left(\cos(\theta_{1(k)} - \theta_{1(k+1)})\right) = L_1 + r$$

$$\Delta\theta_{1(k)} = \theta_{1(k+1)} - \theta_{1(k)} \simeq 0$$

$$\frac{L_1 \left(\theta_{1(k)} - \theta_{1(k+1)}\right)}{\tan(\delta_k)} + L_1 = L_1 + r$$

$$\theta_{1(k+1)} - \theta_{1(k)} = -\frac{r}{L_1} \tan(\delta_k)$$

$$\boxed{\dot{\theta}_1 = -\frac{v}{L_1} \tan(\delta)}$$

Modelo del Robot Truck-Trailer

$$\dot{x} = v \cos(\theta_{12}) \cos(\theta_2)$$

$$\dot{y} = v \cos(\theta_{12}) \operatorname{sen}(\theta_2)$$

$$\dot{\theta}_1 = -\frac{v}{L_1} \tan(\delta) \longrightarrow \dot{\theta}_{12} = -\frac{v}{L_1} \tan(\delta) + \frac{v}{L_2} \operatorname{sen}(\theta_{12})$$

$$\dot{\theta}_2 = -\frac{v}{L_2} \operatorname{sen}(\theta_{12})$$

Fuzzy Control

Human beings are capable of carrying out diverse control activities for making a system behaves in a desired way. Imagine the case of a human driver driving a car for to achieve a desired final destination, avoiding collision with other vehicles, respecting traffic lights, and increasing a decreasing the speed according to traffic conditions, and desired route to follow.

Human beings control the car using a multi-valued logic based on linguistic actions, such as:

- Turn the rudder a little to the left
- Turn the rudder full to the right
- Slowly reduce the car speed
- Speed up quickly
- Keep a safe distance from the car on the right

What is the meaning of “turn the rudder a little to the left”: how many degrees? 5 degrees? 8 degrees? What is the meaning of “slowly reduce the car speed”: to what speed?, 10 m/sec?, 15 m/sec? It is noted that although human beings process vague and imprecise information, they are capable of driving a car safety, precisely, and timely.

Likewise, for driving the car and deciding the rudder angle and speed at each instant of time, human beings do not apply complex mathematical control laws, but they apply a set of rules which have been learned through experience. Usually the rules are of the form IF-THEN meaning that IF a set of conditions are happening, THEN a set of driving actions will be applied.

Fuzzy logic control represent a method for mimicking human control actions for making a system behaves in a desired way, completing a desired task. The set of vague information processed through a set of control rules represent the components of fuzzy logic control which can be applied for stabilization, regulation and tracking control.

Linguistic and Numeric Variables

The variables of a given system could be linguistic variables or numeric variables depending on the type of value assigned to the variables: a linguistic variable has assigned linguistic values, a numeric variable has assigned numeric values.

When we say “turn the rudder a little to the left” the variable is the steering angle δ and the assigned value is “a little to the left”:

steering angle δ = a little to the left

In this case, the steering angle has assigned a linguistic value and, therefore, it is a linguistic variable.

When we say “turn the rudder 15 degrees to the left, the variable steering angle δ is assigned the value +15 degree:

steering angle δ = +15 degrees

In this case, the steering angle has assigned a numeric value and, therefore, it is a numeric variable.

By their own nature linguistic variables are vague, fuzzy and diffuse, but numeric variables are precise and can be represented with the required level of accuracy. In the following more examples:

Today the temperature is high (linguistic variable)
Today the temperature is 32 ^{O}C (numeric variable)

The heart rate is very low (linguistic variable)
The heart rate is 45 beats per second (numeric variable)

Mary is very tall (linguistic variable)
The height of Mary is 1.88 m (numeric variable)

A linguistic variable could have assigned a linguistic within a set of possible linguistic values. In the following, the variable temperature could be assigned several linguistic values

Temperature is extremely low
The temperature is very low
Temperature is low
Temperature is mild
Temperature is high
Temperature is very high
Temperature is extremely high

Linguistic and numeric values are related, usually with many numeric values corresponding to the same linguistic variable:

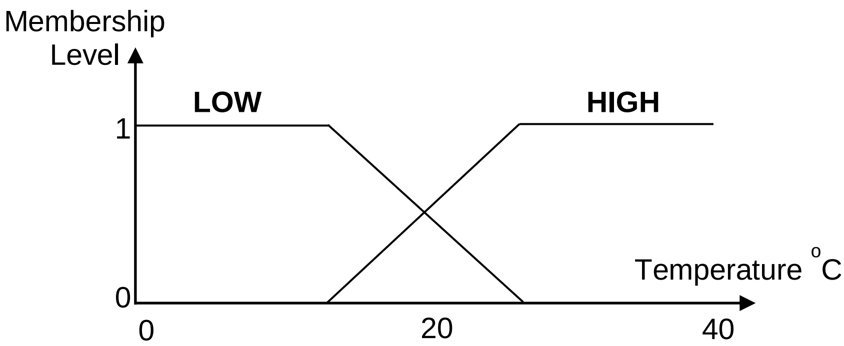
The temperature is 36^{O}C, it is very high
The temperature is 35^{O}C, it is very high
The temperature is 34^{O}C, it is very high

The temperature is 33 ^{O}C, it is high
The temperature is 32 ^{O}C, it is high
The temperature is 31 ^{O}C, it is high

Also, a temperature of 33 ^{O}C is more "high" than a temperature of 31 ^{O}C. Even more, a temperature of 33^{O}C could be considered "very high" for some people and "high" for another people. In other words, a numeric could belong to different linguistic values but with different membership level. This particular relationship between linguistic and numeric values is represented by the membership function.

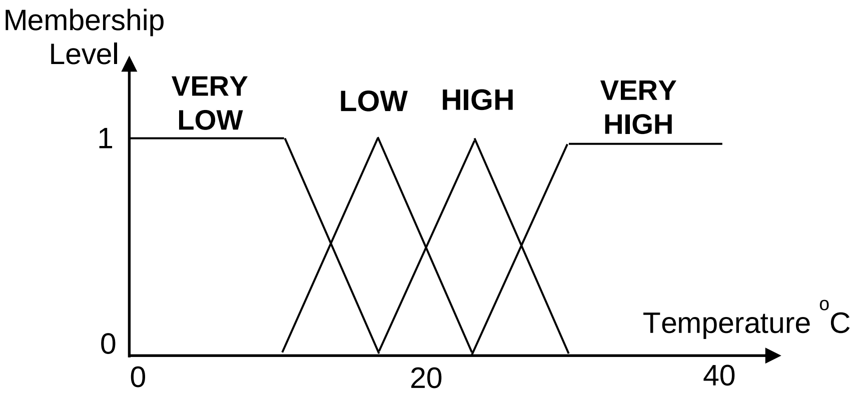
Membership Function

A membership function is defined for each linguistic value of a variable. It represents the level of membership of a numeric value to a linguistic value. Considering the variable temperature is in the range [0^{0}C 40^{0}C], and it has defined only two linguistic values (Low and High), then the membership function defined for each linguist value could be as it is shown in Figure.



In Figure it is noted that temperatures close to 0^{o}C have a membership level equal to 1 for linguistic value "Low" but they have a membership level equal to 0 for linguistic value "High". Likewise, temperatures close to 40^{o}C have a membership level equal to 1 for linguistic value "High" but they have a membership level equal to 0 for linguistic value "Low". However, temperatures around 20^{o}C have membership levels belonging to both linguistic values "Low" and "High".

Considering the variable temperature is in the range [0^{0}C 40^{0}C], and it has defined four linguistic values (Very Low, Low, High, and Very High), then the membership function of each linguist value could be as it is shown in Figure.



In Figure it is noted that temperatures close to 0 °C have a membership level equal to 1 for linguistic value “Very Low” but they have a membership level equal to 0 for linguistic values “Low”, “High” and “Very High”. Likewise, temperatures close to 40 °C have a membership level equal to 1 for linguistic value “Very High” but they have a membership level equal to 0 for linguistic values “Very Low”, “Low” and “High”.

In the Figure it is noted that there is a temperature belonging to linguistic value “High” with a membership level equal to 1, and that the membership level decreases for higher and smaller temperature numeric values. Instead of triangular membership functions for linguistic values “Low” and “High”, trapezoidal functions can be considered. The important point is the membership function reflects the membership level of a numeric value to a linguistic value.

Membership functions are defined so that there is an overlapping between two contiguous partitions so that there is always defined a membership function for each numeric value. Although it is not a mandatory requirement, overlapping is considered only for two adjacent membership functions. Likewise, membership functions are defined so that the sum of the membership values of a numeric value to each membership function is one.

IF-THEN Rules

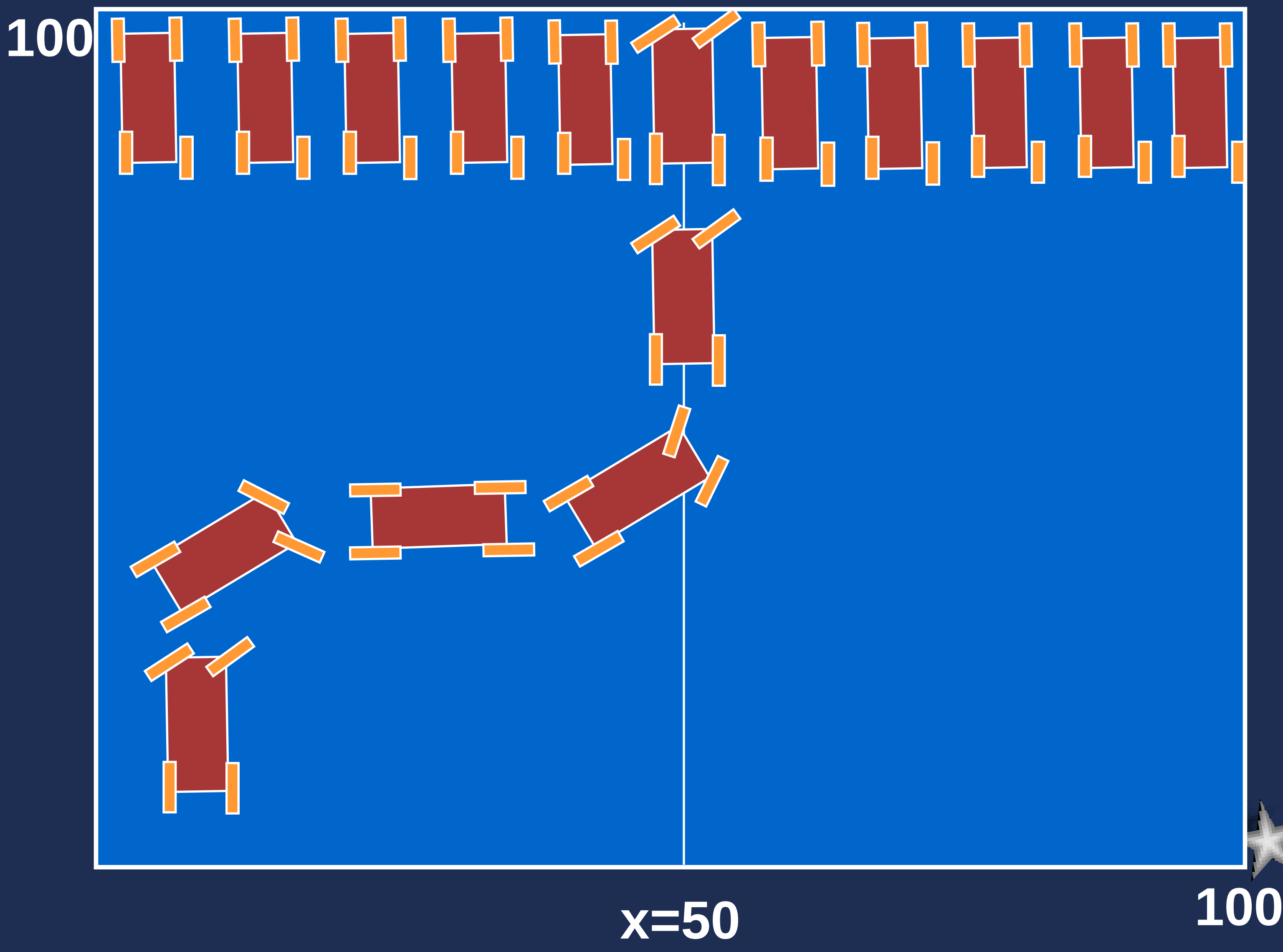
IF-THEN rules are fuzzy logic statements used to formulate the conditional relationships between linguistic variables. The IF part is called the “antecedent” and the THEN part is called the “consequent”. In fuzzy logic control, an IF-THEN rule states what the control action should be when the antecedent variables have given linguistic values. In the following there are some examples of IF-THEN rules:

IF temperature is “Very Low” THEN turn the heater on and set a “High” heating temperature
IF heart rate is “High” THEN take a “Long” rest
IF heart rate is “Low” AND blood pressure is “Low” THEN take a “High” doses of medicine
IF traffic is “Low” THEN increase speed to “High” and keep at the “Middle” of the lane

A fully logic control system comprises several IF-THEN rules defined to cover all possible combinations or linguistic values of the variables in the Antecedent part. The complete set of IF-THEN rules is called the Rule Base.
Each IF-THEN rule is generated considering the knowledge and experience of a human expert who knows what to do in the case the Antecedent part is active. The Rule Base represents the core of a fuzzy control system and it comprises the knowledge and experience of the human expert required for solving the control problem.

Fuzzy Control

Control Difuso de Robot Móvil Tipo Carro

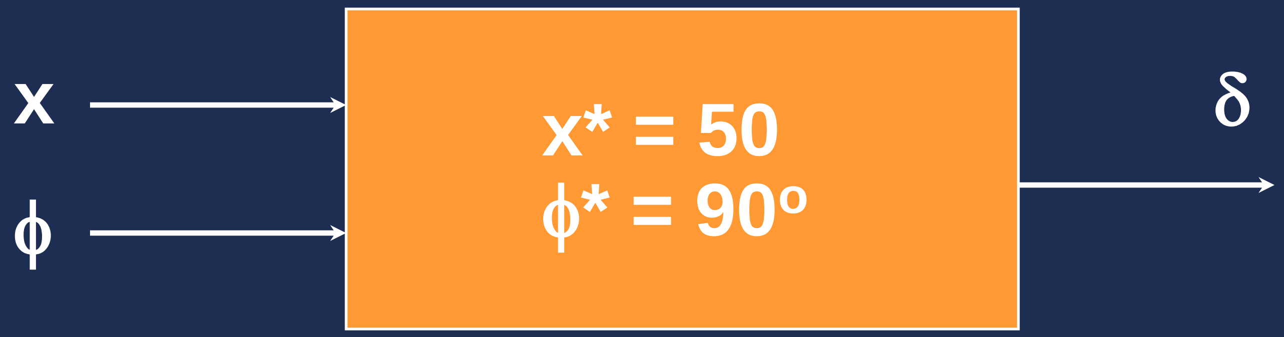


$x^* = 50$

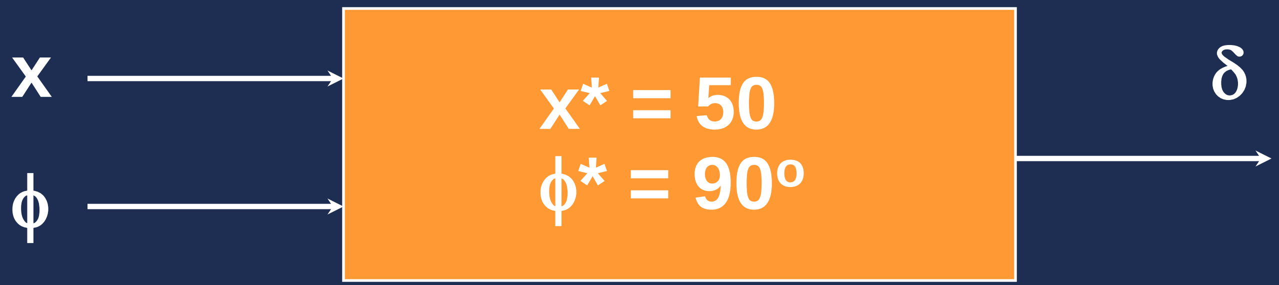
$y^* = 100$

$\phi^* = 90^o$

Controlador Difuso

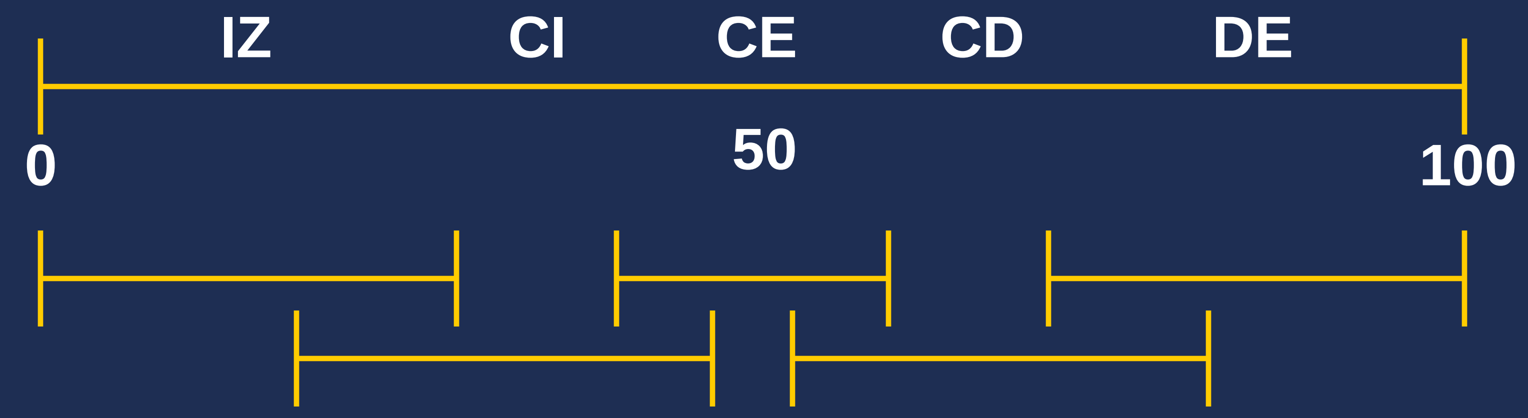


Diseño de Controlador Difuso

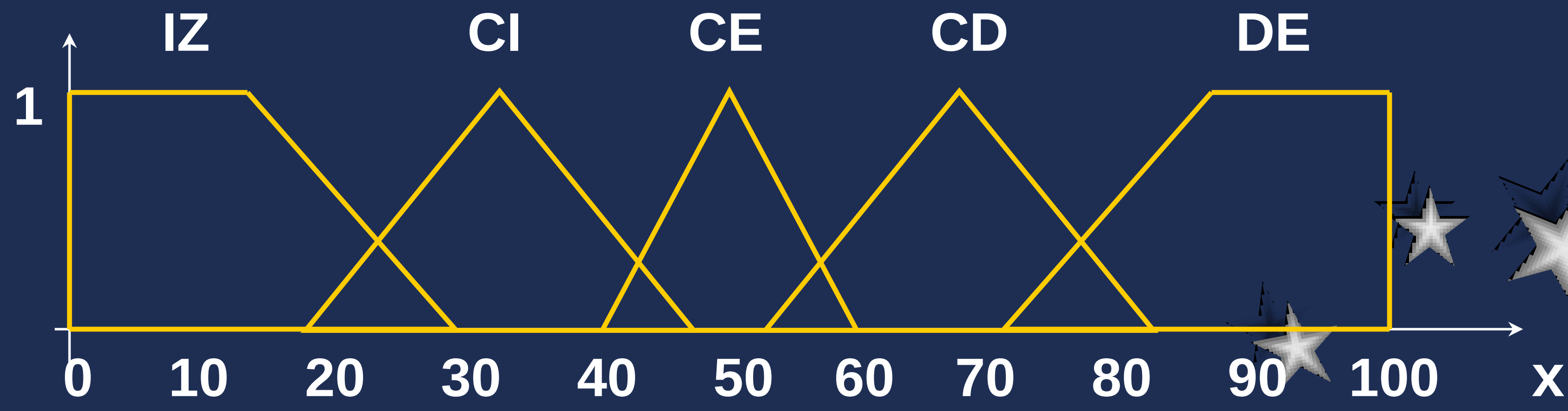


- Determinar rango de variación de x, ϕ, δ
- Particionar cada rango de variación
- Asignar una variable linguística a cada partición
- Definir funciones de pertenencia para cada partición
- Determinar base de reglas (experiencia humana)

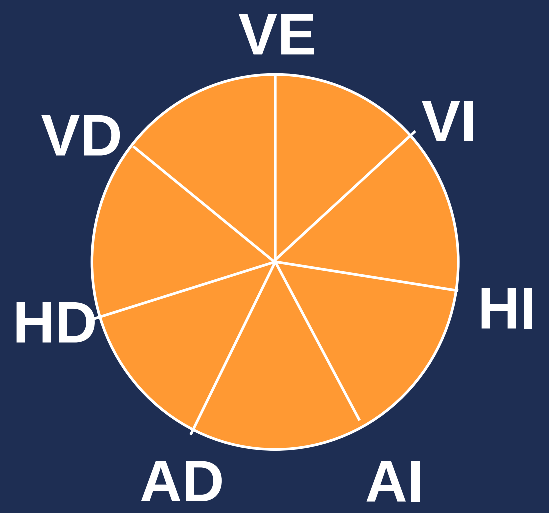
Particiones de x

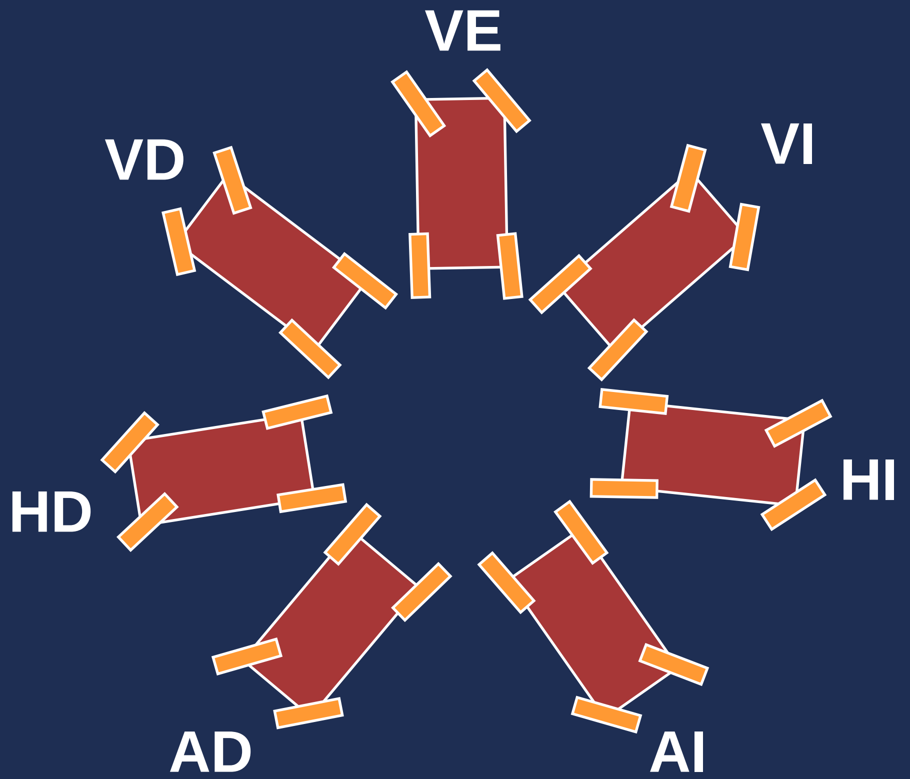


Funciones de Pertenecia de x

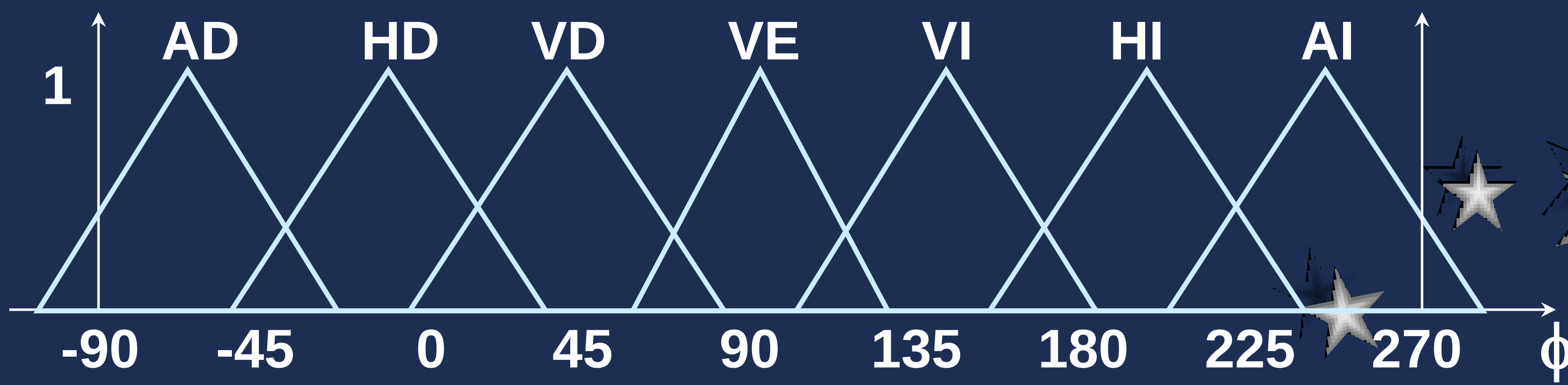


Particiones de ϕ



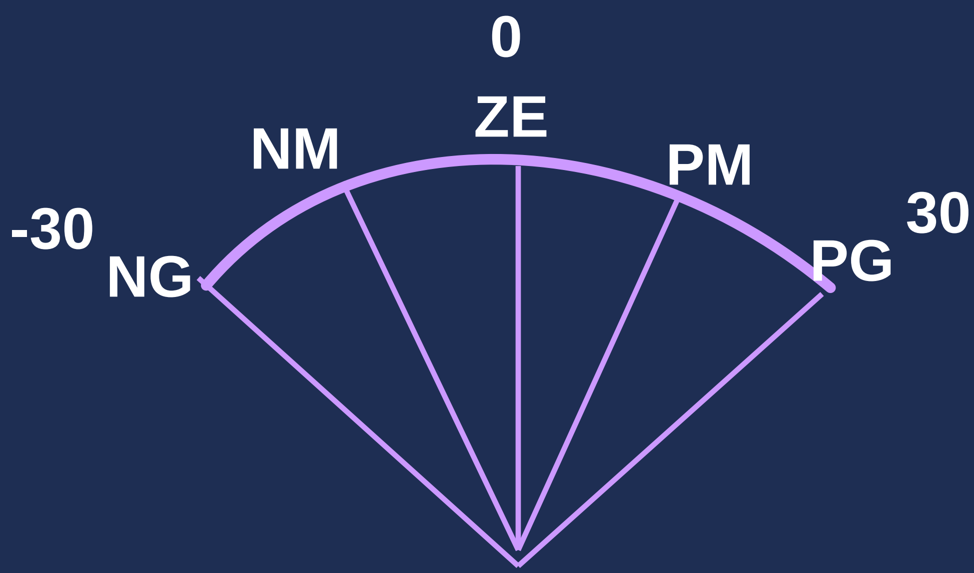


Funciones de Pertenencia de ϕ

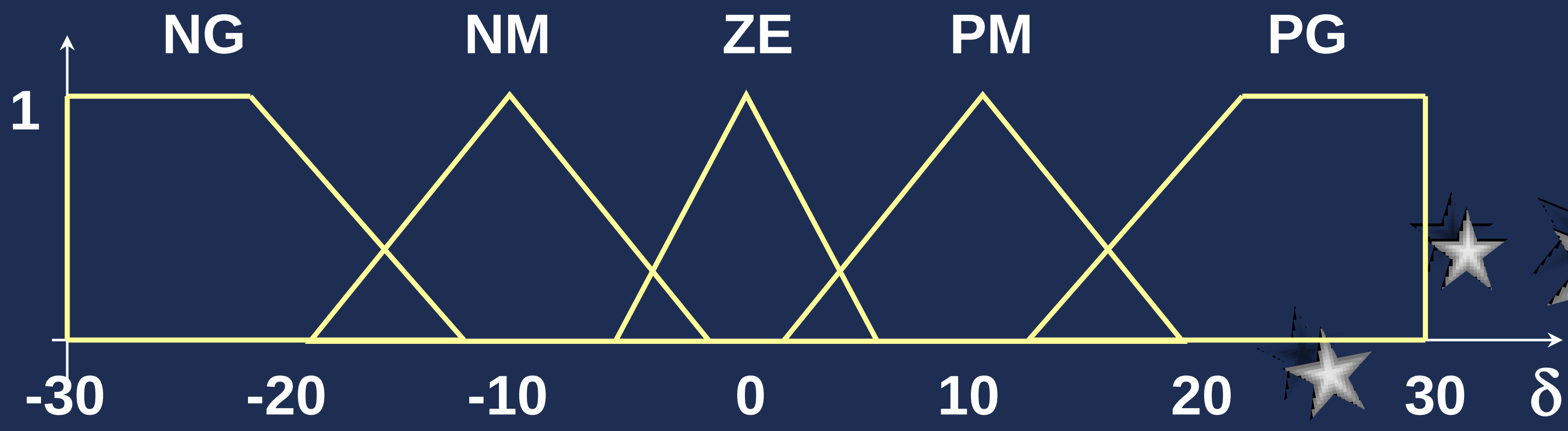


Particiones de δ

$-30^o <= \delta <= 30^o$

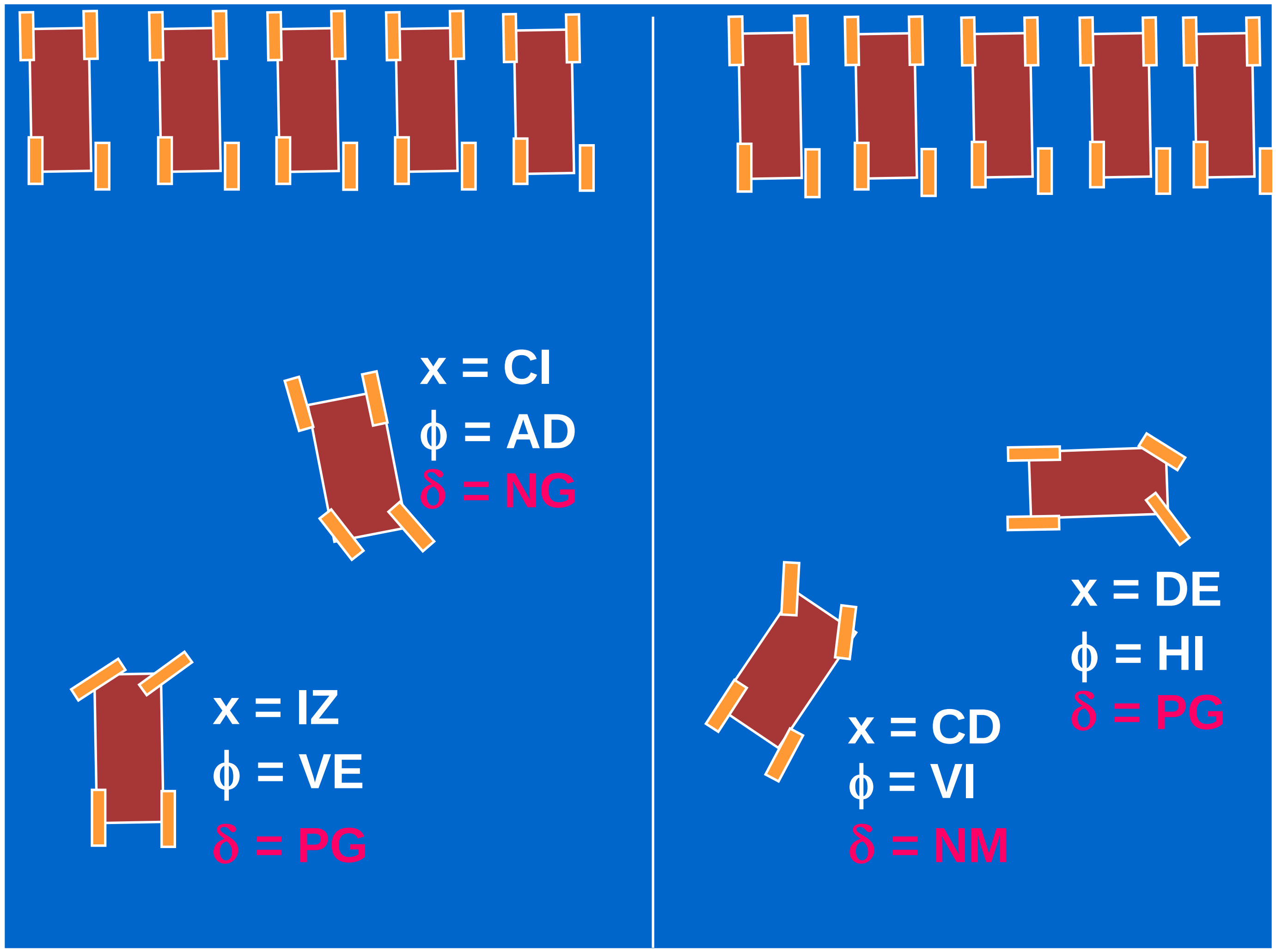


Funciones de Pertenecia de δ



Base de Reglas

	IZ	CI	CE	CD	DE
AD		NG			
HD					
VD					
VE	PG				
VI				NM	
HI					PG
AI					

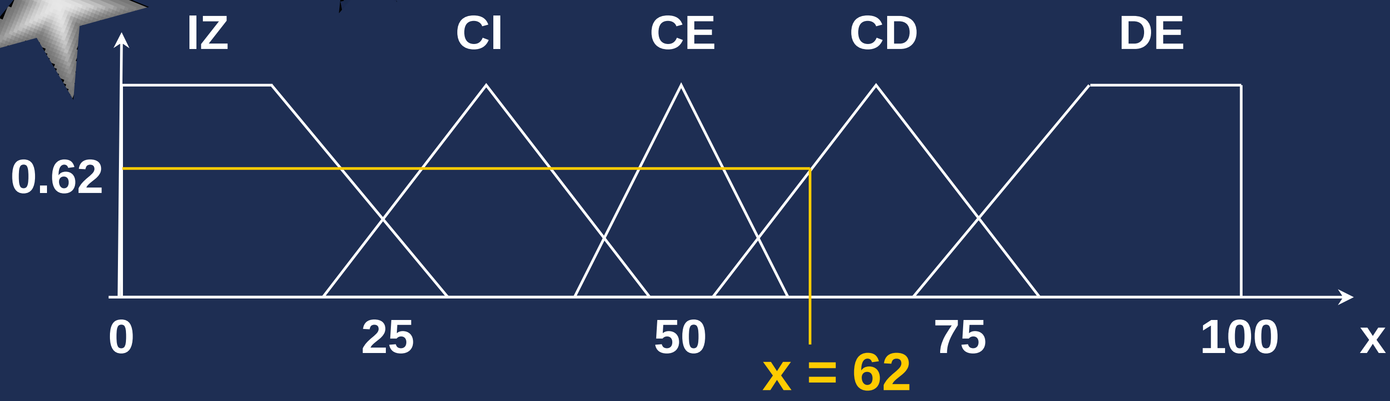


x=50

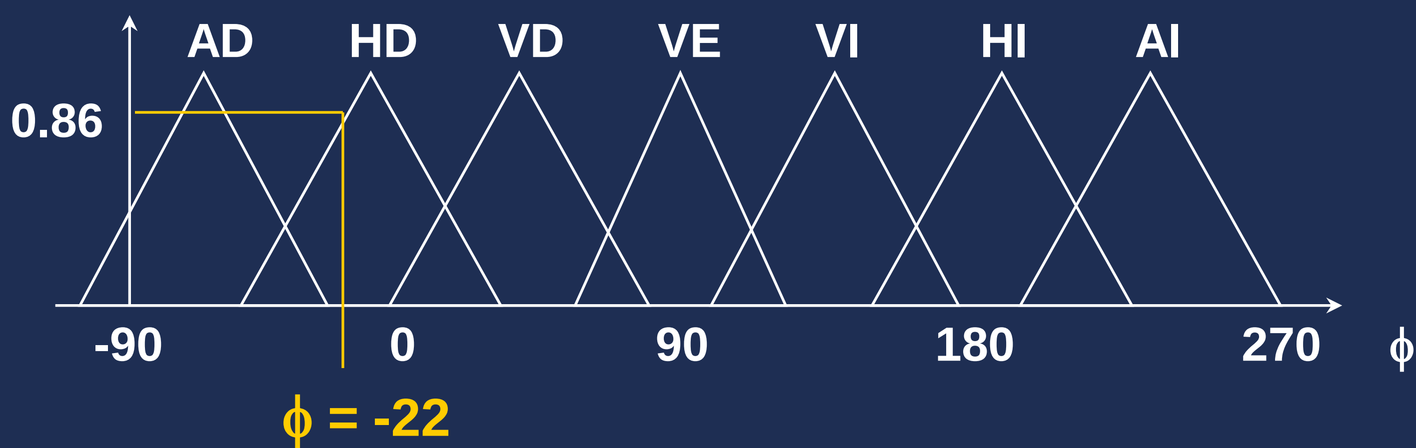
Una vez diseñado el controlador difuso se usa en tiempo real para controlar el robot móvil

Dado un valor de *x* y un valor de ϕ, se calcula el valor de δ usando el método de inferencia *min-max*

x = 62
ϕ = -22



x = CD



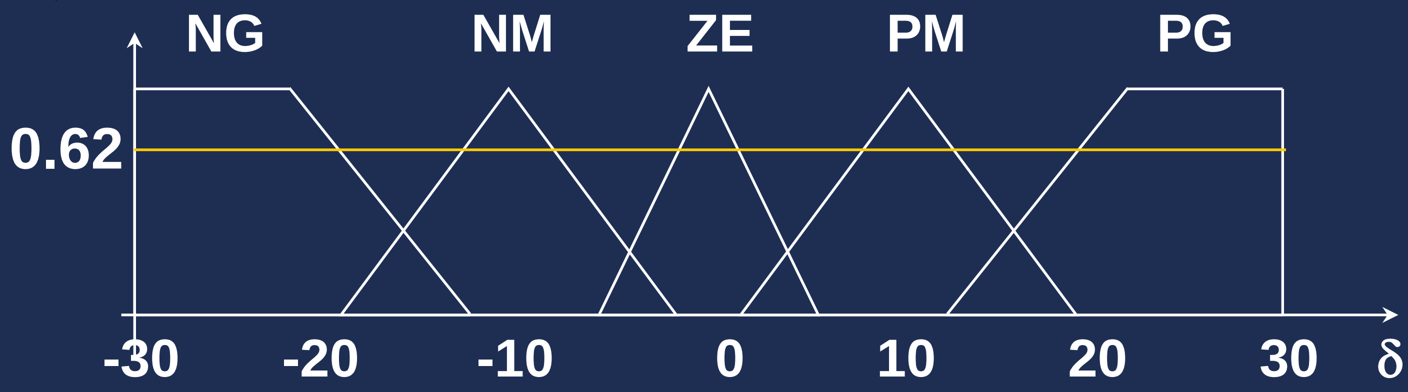
ϕ = HD

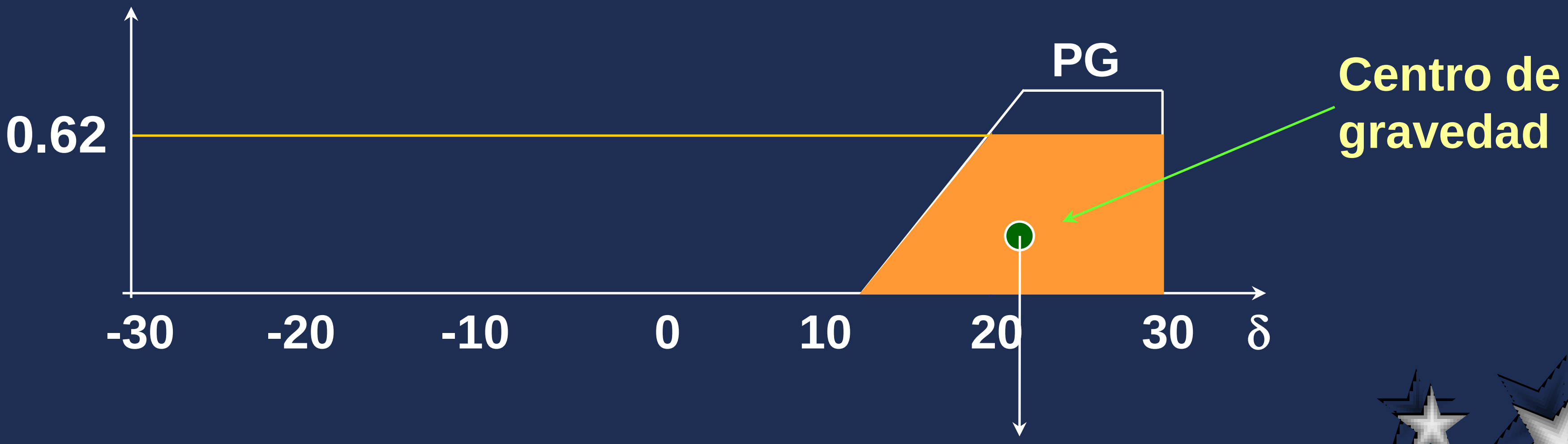
x = CD y ϕ = HD $\longrightarrow$ δ = PG

min(0.62, 0.86) = 0.62

δ = PG --- 0.62

δ = PG --- 0.62

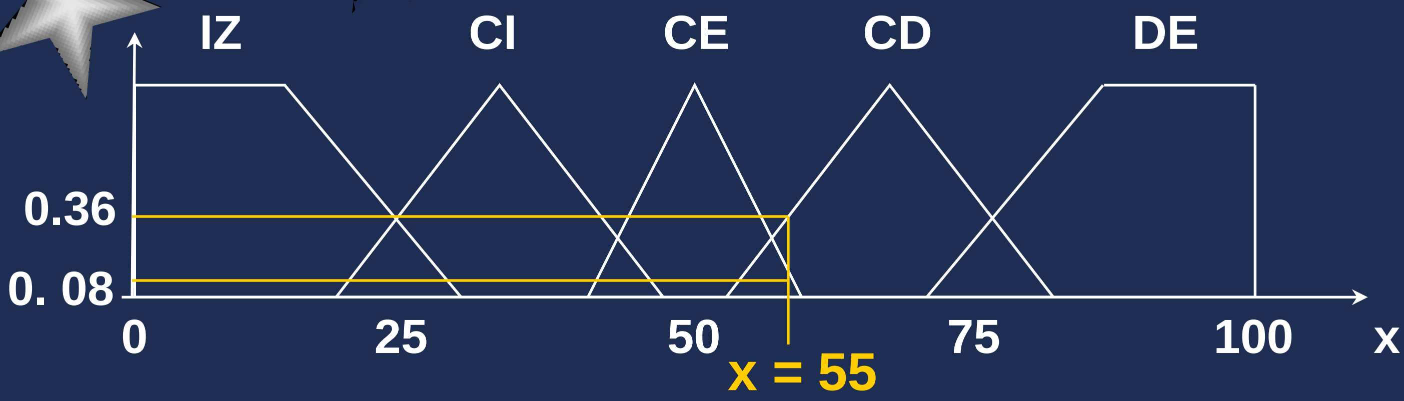




δ = 22.5º

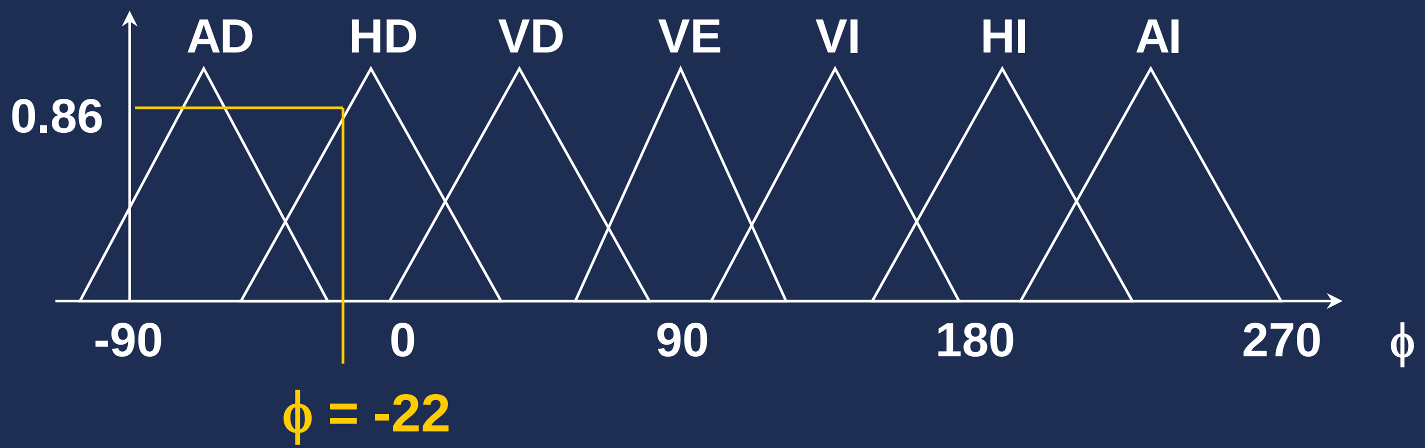
$x = 55$

$\phi = -22$



$x = CE$

$x = CD$



$\phi = HD$

$x = CE$ y $\phi = HD \longrightarrow \delta = NG$

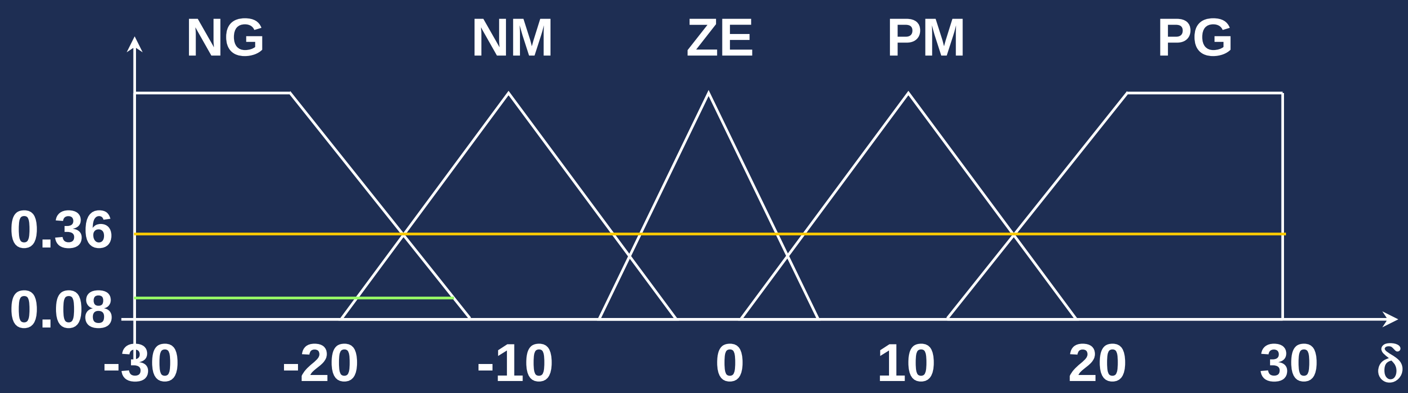
$x = CD$ y $\phi = HD \longrightarrow \delta = PG$

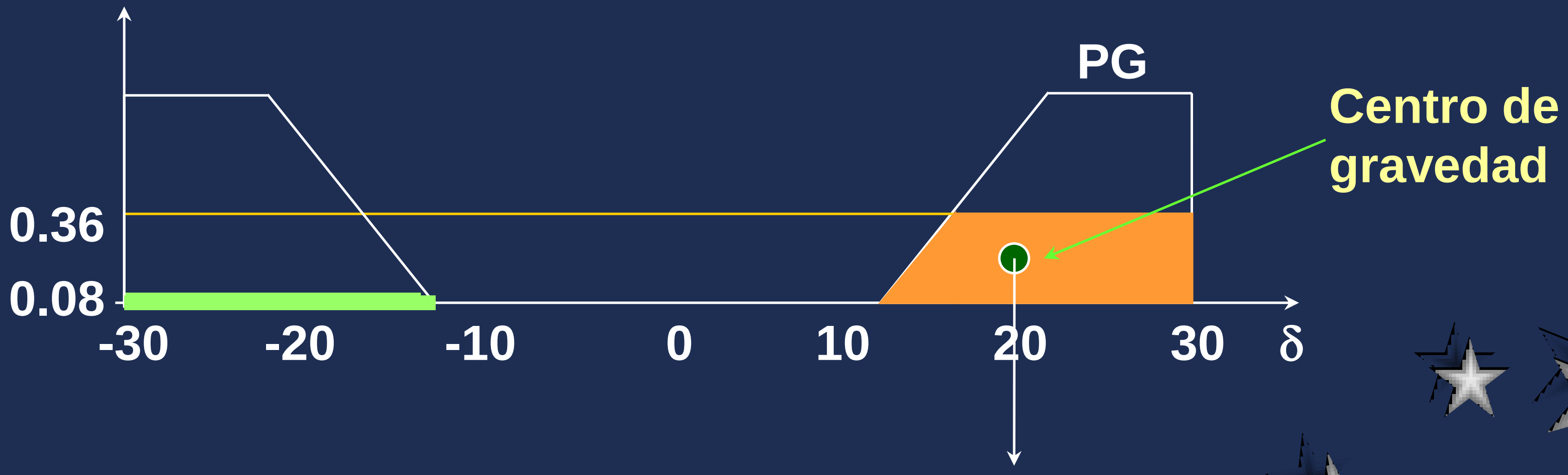
min(0.08, 0.86) = 0.08

min(0.36, 0.86) = 0.36

δ = NG --- 0.08

δ = PG --- 0.36

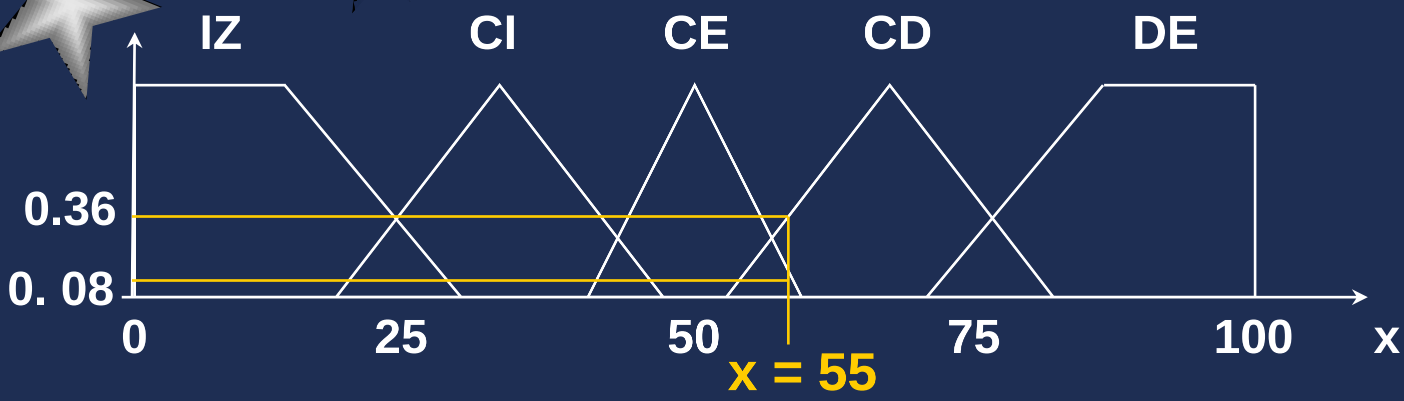




δ = 20.2º

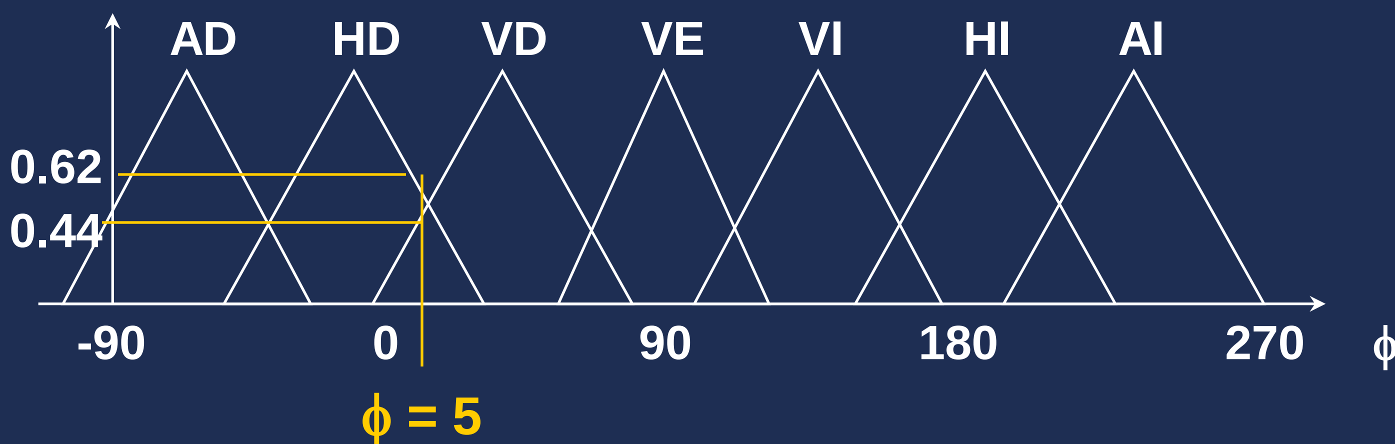
$x = 55$

$\phi = 5$



$x = CE$

$x = CD$



$\phi = HD$

$\phi = VD$

$x = CE$ y $\phi = HD \longrightarrow \delta = NG$ min(0.08, 0.62) = 0.08

$x = CE$ y $\phi = VD \longrightarrow \delta = PG$ min(0.08, 0.44) = 0.08

$x = CD$ y $\phi = HD \longrightarrow \delta = PG$ min(0.36, 0.62) = 0.36

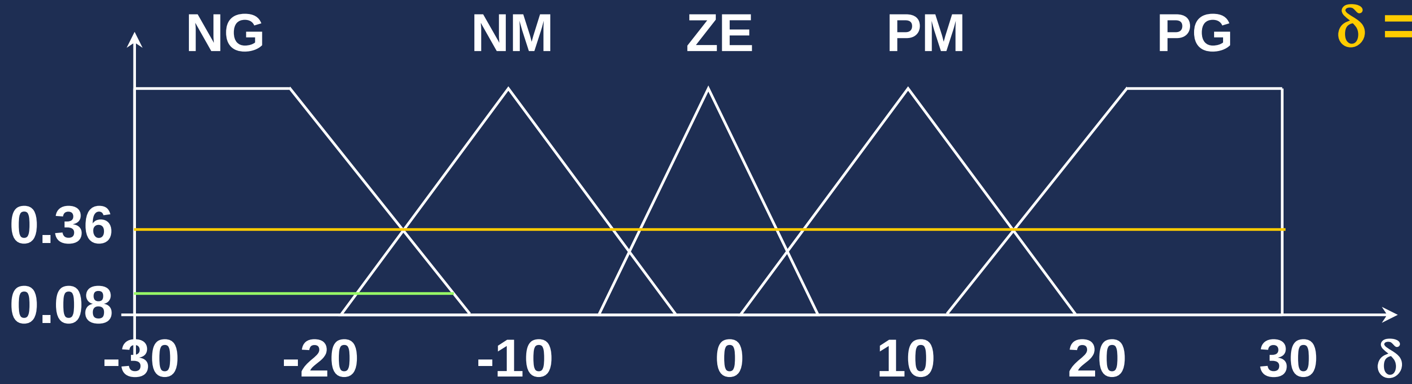
$x = CD$ y $\phi = VD \longrightarrow \delta = PM$ min(0.36, 0.44) = 0.36

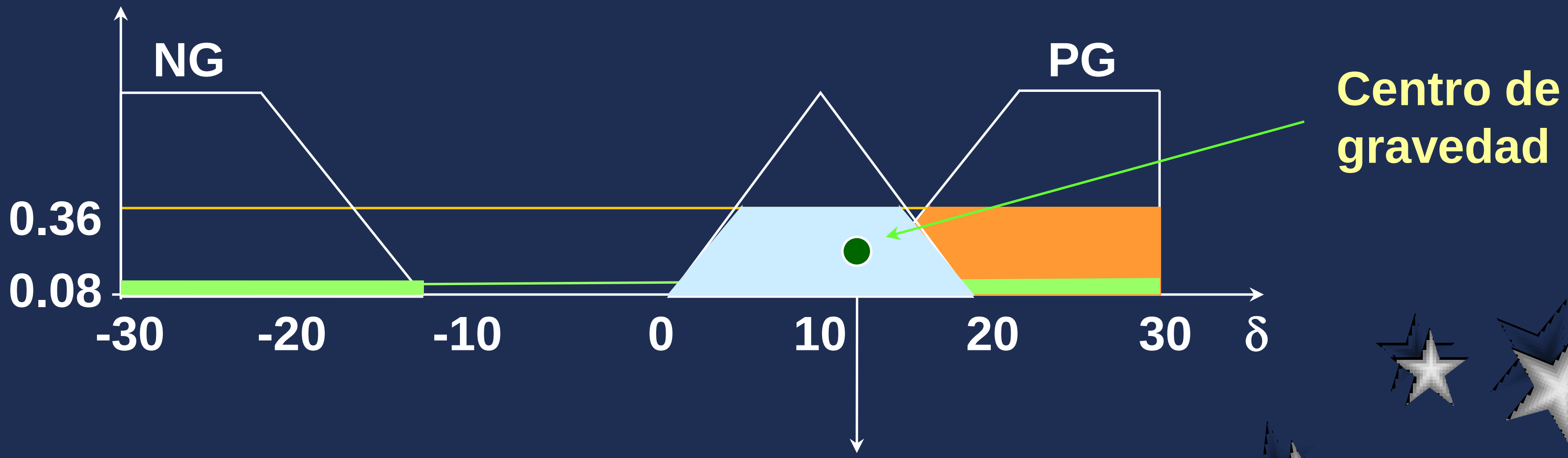
δ = NG --- 0.08

δ = PG --- 0.08

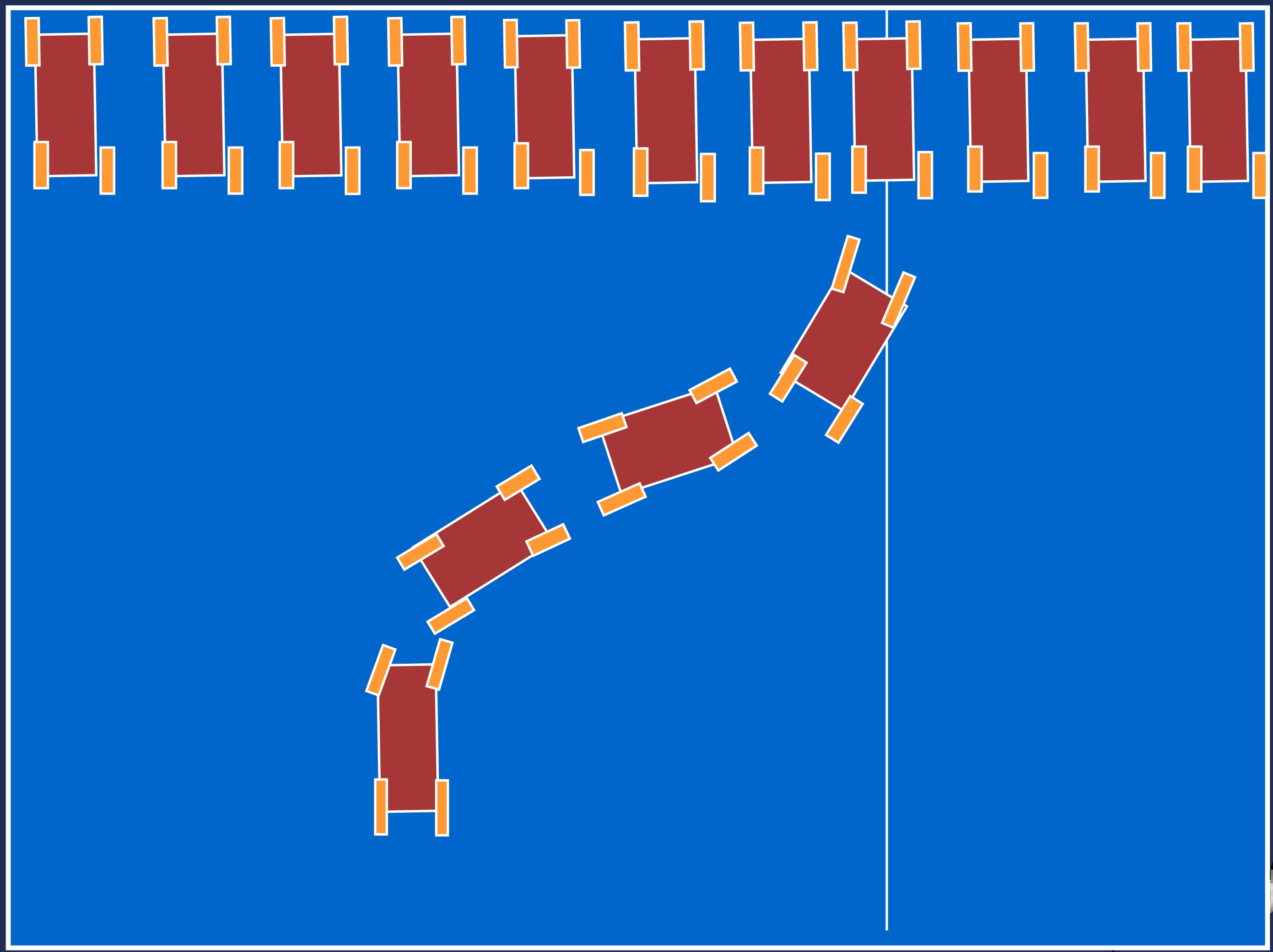
δ = PG --- 0.36

δ = PM --- 0.36





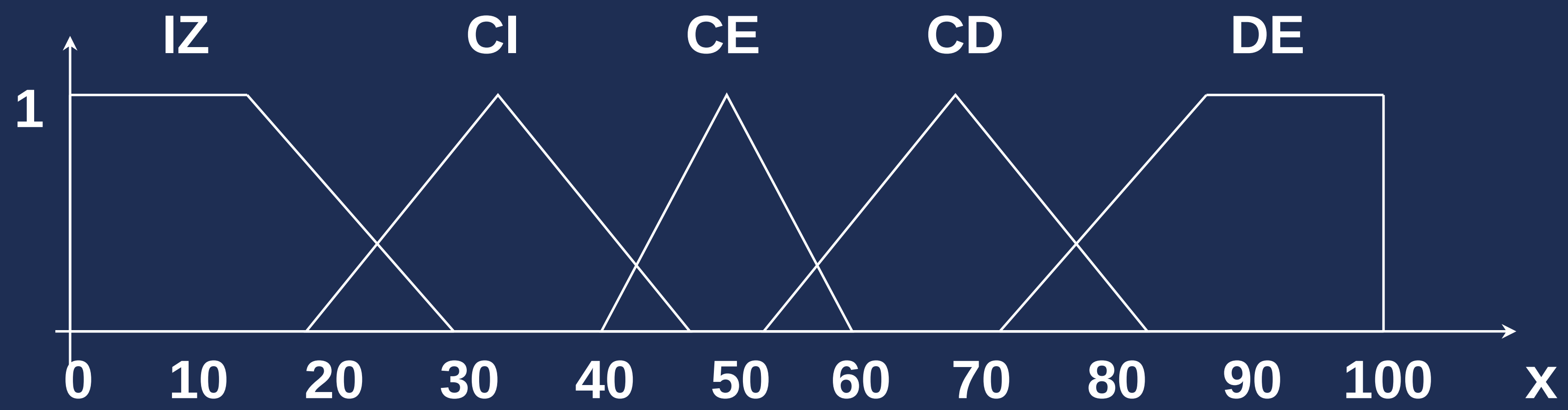
δ = 12.6º



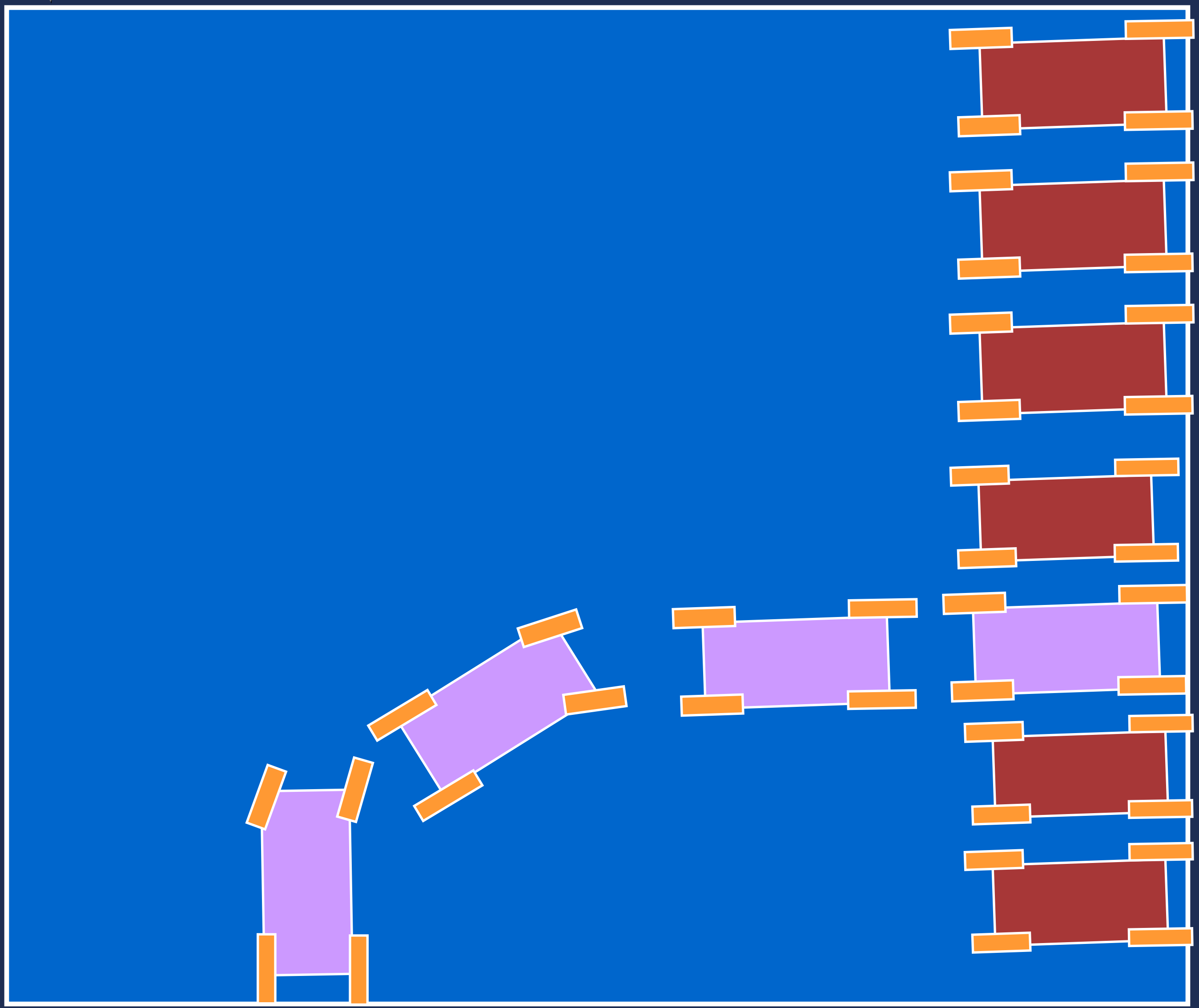
x=70

Otra Posición Deseada x*

Funciones de Pertenecia de x

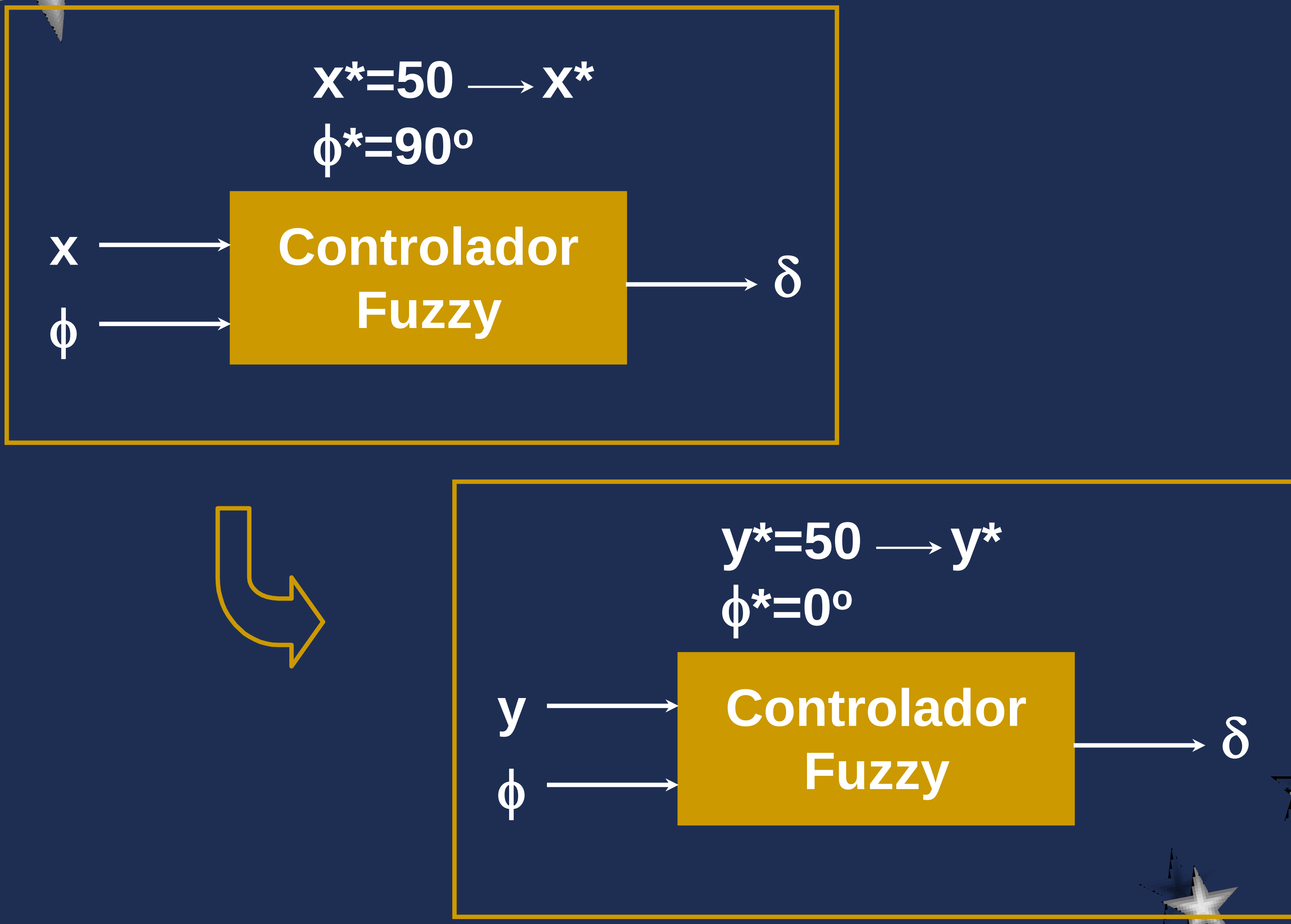


x = x - (x* - 50)

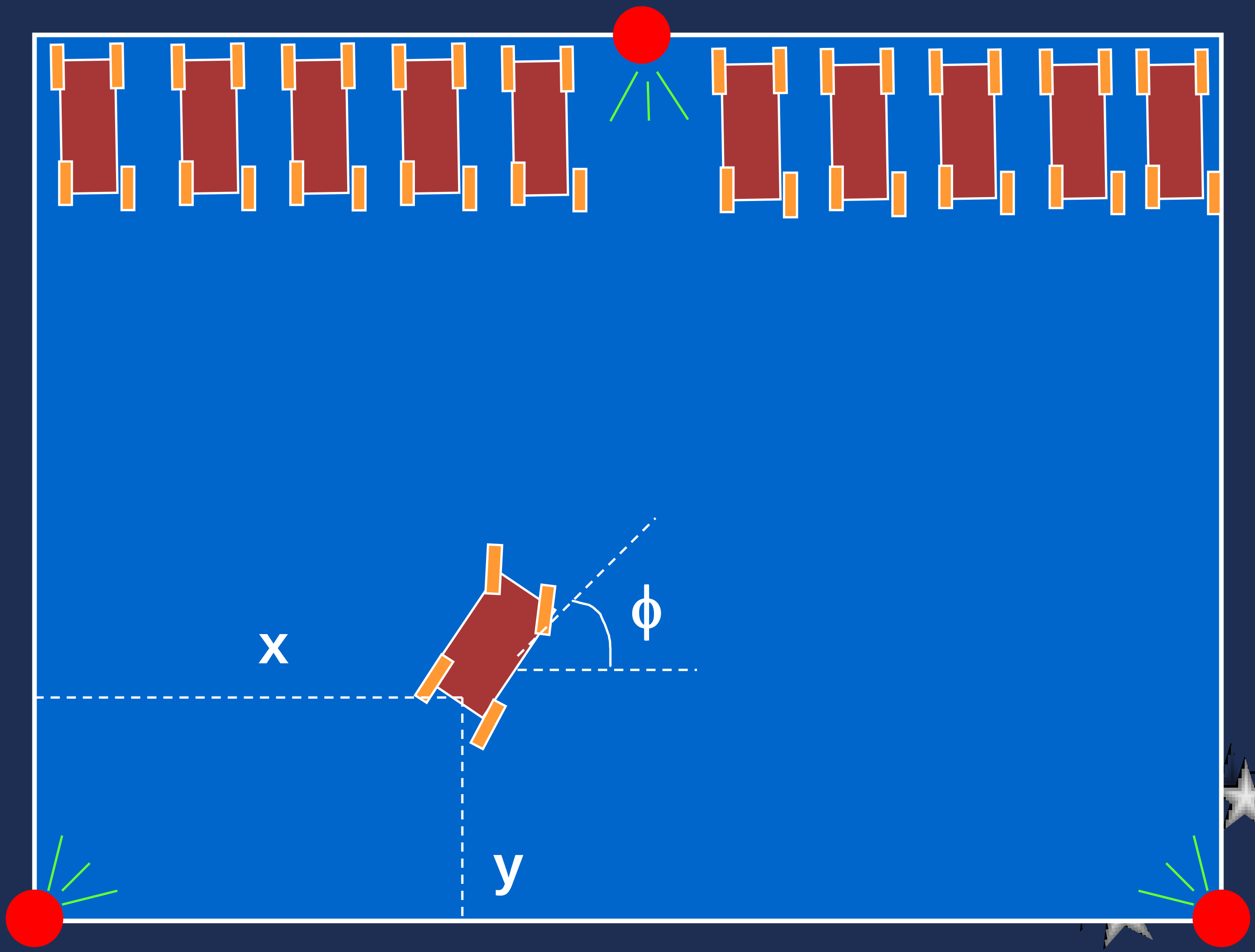


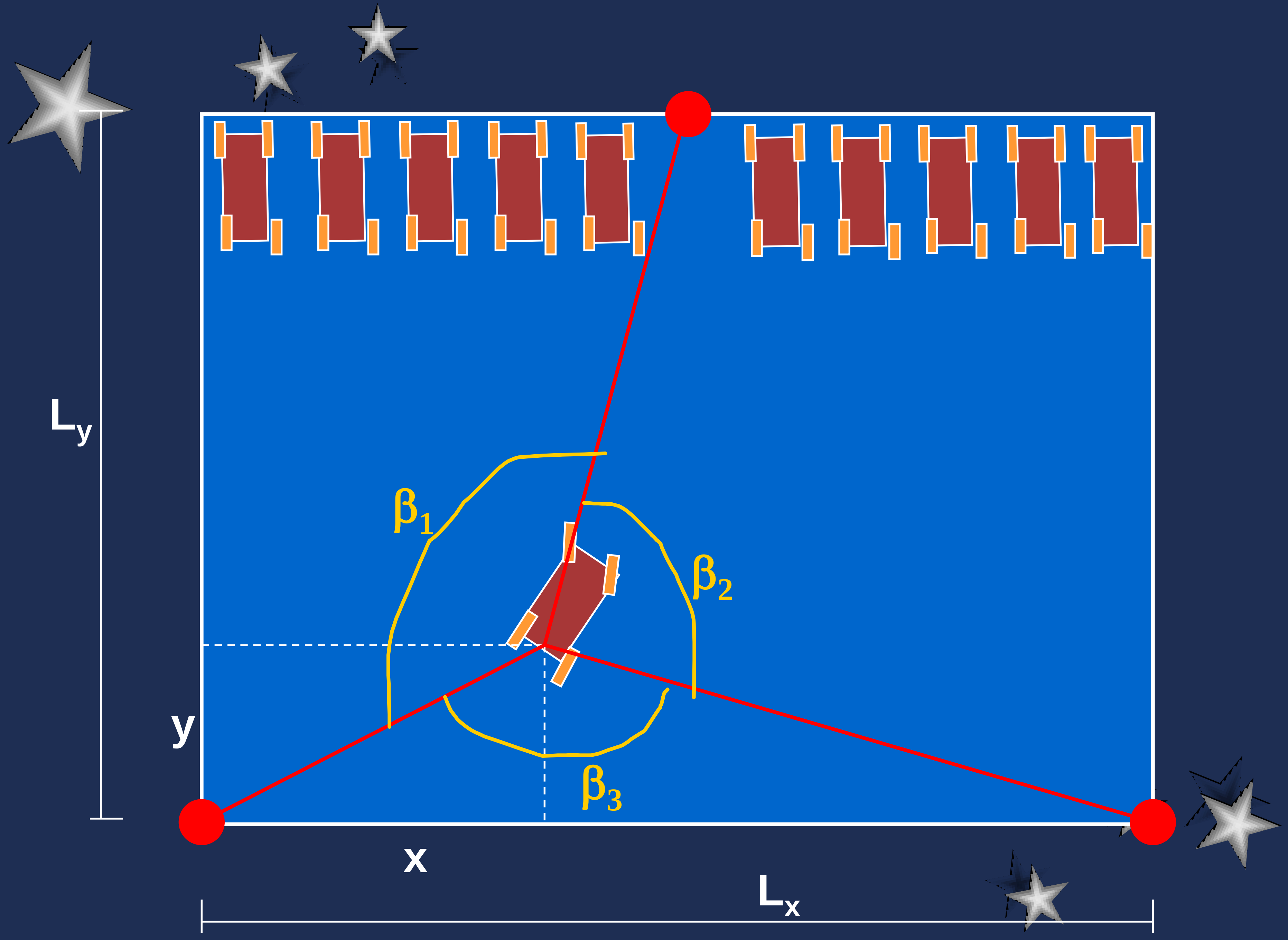
y*=30

ϕ*=0


x*=50 → x*
ϕ*=90º
x
ϕ
Controlador Fuzzy
δ
y*=50 → y*
ϕ*=0º
y
ϕ
Controlador Fuzzy
δ

Problema de Navegacion





L_y
β_1
β_2
β_3
y
x
L_x